

Process Dynamics & Control

Process Dynamics & Control (CH 61016)

Class Teacher:

1. Prof Amar Nath Samanta
2. Prof Amiya K Jana

Class TA

Mr. Shubham Raj
Mr. Vishal Kamble

Mr. Guru Bhattacharya

Class Timing

Wednesday : 10 AM to 11 AM
Thursday : 9 AM to 10 AM
Friday : 11 AM to 1 PM

Attendance

Biometric method
Institute Deregistration Rule

Syllabus of PDC (my part only)

- Development Process Models
- Development of Control Relevant Models
- Review of linear Dynamics
- Development of controller and observer using state space models
- Non-linear system analysis
 - Phase Plane
 - Bifurcation
 - Stability analysis
- Feedback linearization of nonlinear systems

Process Dynamics & Control

Books:

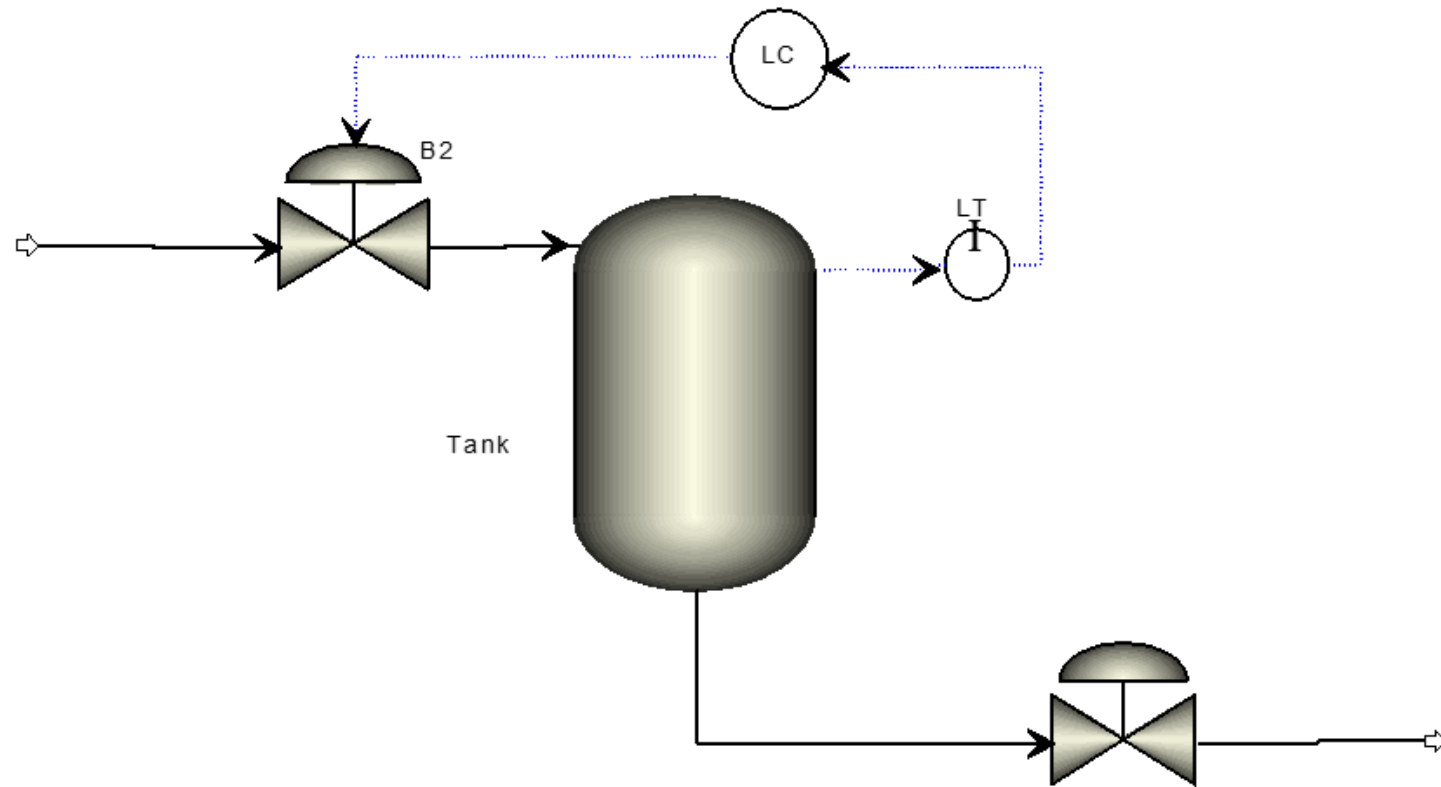
1. Process Dynamics & Control -- Seborg, Edgar, Mellichamp & Doyle
2. Process Control -- W Bequette
3. Process Dynamics, Modelling & Control
-- B A Ogunnaike & W H Ray
4. Modern Control Engineering -- K Ogata

Software: Knowledge of MATLAB & Simulink desirable

Why study Process dynamics

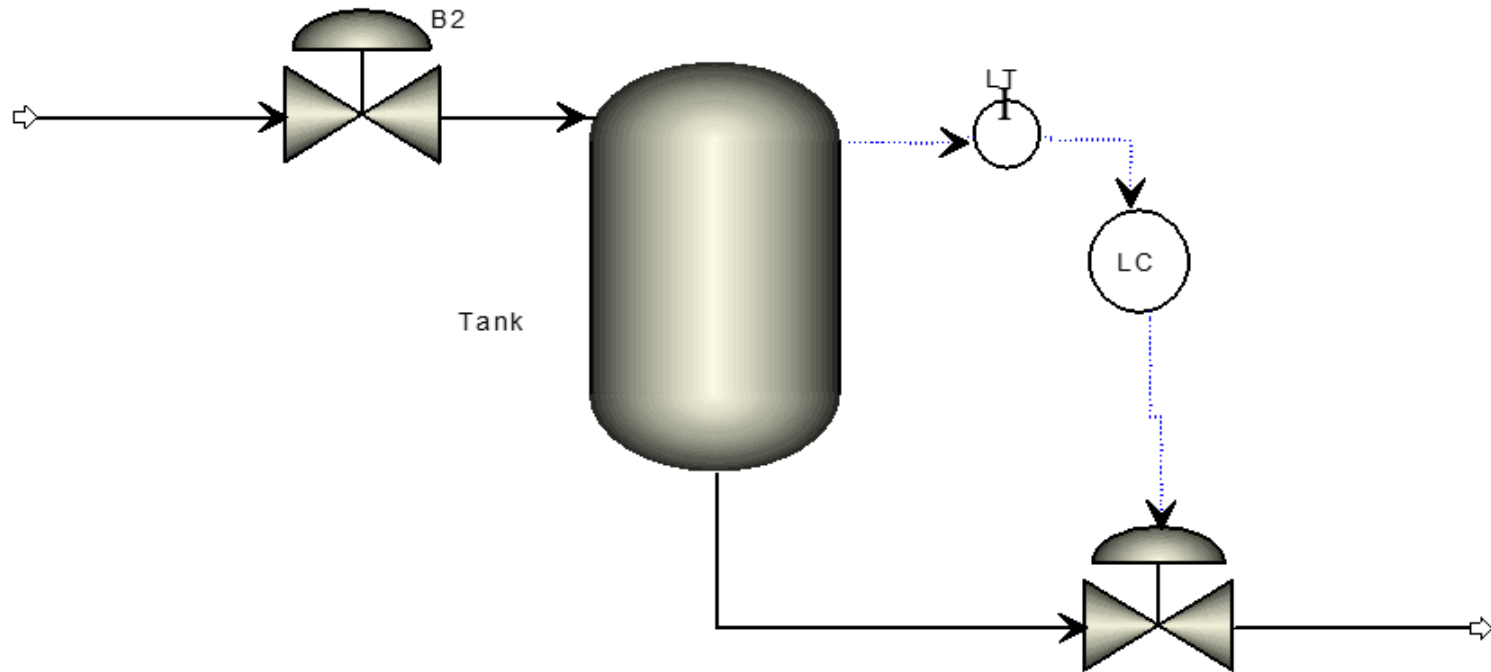
Let us consider a very simple example of tank level control problem

Option 1:



Tank Level problem

Option 2:



Which configuration is better from the control view point and why?

Benchmark Systems to be used for the course

- Tank Level System
 - 3 tank in series
 - Quadruple Tank
- Continuous Stirred Tank Heater
- Continuous Stirred Tank Reactor system
 - Isothermal Reactor (Van de Vusse reaction)
 - Non-isothermal Reactor
 - Polymerization Reactor
- Distillation Column

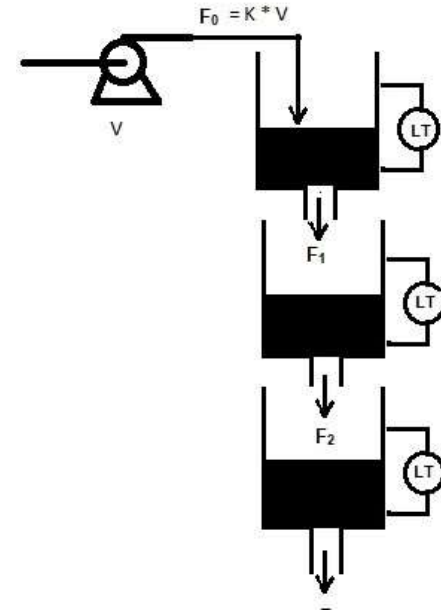
MATLAB / SIMULINK model will be developed based on nonlinear model

Mathematical Model of tank level system

$$\frac{dh_1}{dt} = \frac{F_0}{A_1} - \frac{a_1}{A_1} \sqrt{2gh_1}$$

$$\frac{dh_2}{dt} = \frac{a_1}{A_2} \sqrt{2gh_1} - \frac{a_2}{A_2} \sqrt{2gh_2}$$

$$\frac{dh_3}{dt} = \frac{a_2}{A_3} \sqrt{2gh_2} - \frac{a_3}{A_3} \sqrt{2gh_3}$$



Control objective:

Control the 3rd tank level by manipulating pump voltage V , where $F_0 = K V$

Add q_2 input flow on the tanks 2 and set q_2 as manipulated variable.

Which one is difficult to control?

Data for Simulation in Matlab/Simulink

$$A_1, A_3 = 28 \text{ cm}^2 \quad A_2 = 32 \text{ cm}^2 \quad V^s = 3 \text{ V}$$

$$a_1 = 0.06725 \text{ cm}^2 \quad a_2 = 0.05683 \text{ cm}^2$$

$$a_3 = 0.07089 \text{ cm}^2 \quad K=3.14$$

- Transfer Function model

Define : $x_1 = h_1 - h_1^s$; $x_2 = h_2 - h_2^s$; $x_3 = h_3 - h_3^s$; $u = V - V^s$

Linearizing around steady state value h_1^s

$$\frac{dh_1}{dt} = \frac{K}{A_1} V - \frac{a_1}{A_1} \sqrt{2g} \left[\sqrt{h_1^s} + \frac{1}{2\sqrt{h_1^s}} (h_1 - h_1^s) \right]$$

Subtracting above from steady state equation,

$$\frac{dx_1}{dt} = \frac{K}{A_1} u - \frac{a_1 \sqrt{g}}{A_1 \sqrt{2h_1^s}} x_1 \quad \rightarrow \text{Transfer function } \frac{x_1(s)}{u(s)} = \frac{K_{p1}}{\tau_{p1}s+1}$$

$$\text{Where, } K_{p1} = \frac{K}{a_1} \sqrt{\frac{2h_1^s}{g}} \quad \text{and } \tau_{p1} = \frac{A_1}{a_1} \sqrt{\frac{2h_1^s}{g}}$$

Similarly,

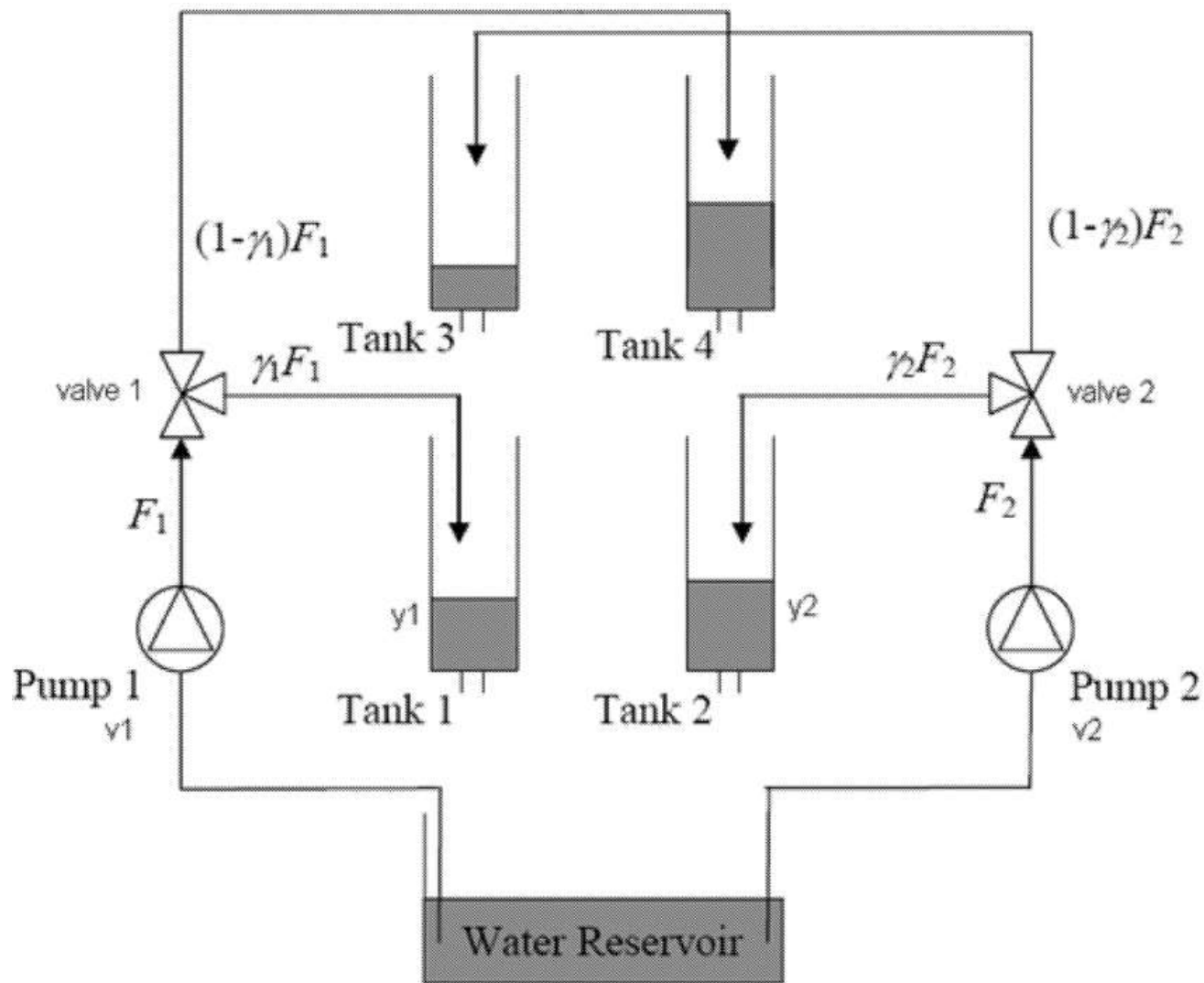
$$\frac{dx_2}{dt} = \frac{a_1}{A_2} \sqrt{\frac{g}{2h_1^s}} x_1 - \frac{a_2}{A_2} \sqrt{\frac{g}{2h_2^s}} x_2 \quad \text{and Transfer function } \frac{x_2(s)}{x_1(s)} = \frac{K_{p2}}{\tau_{p2}s+1}$$

$$\frac{dx_3}{dt} = \frac{a_2}{A_3} \sqrt{\frac{g}{2h_2^s}} x_2 - \frac{a_3}{A_3} \sqrt{\frac{g}{2h_3^s}} x_3 \quad \text{and Transfer function } \frac{x_3(s)}{x_2(s)} = \frac{K_{p3}}{\tau_{p3}s+1}$$

$$\text{Where, } K_{pj} = \frac{a_{j-1}}{a_j} \sqrt{\frac{h_j^s}{h_{j-1}^s}} \quad \text{and } \tau_{pj} = \frac{A_j}{a_j} \sqrt{\frac{2h_j^s}{g}} \quad \text{for } j = 2, 3$$

$$\text{So, Process Transfer function } G(s) = \frac{x_3(s)}{u(s)} = \frac{K_{p1} K_{p2} K_{p3}}{(\tau_{p1}s+1)(\tau_{p2}s+1)(\tau_{p3}s+1)}$$

Quadruple Tank problem



Quadruple Tank problem

$$\begin{aligned}\frac{dh_1}{dt} &= -\frac{a_1}{A_1} \sqrt{2gh_1} + \frac{a_3}{A_1} \sqrt{2gh_3} + \frac{\gamma_1 k_1}{A_1} v_1 \\ \frac{dh_2}{dt} &= -\frac{a_2}{A_2} \sqrt{2gh_2} + \frac{a_4}{A_2} \sqrt{2gh_4} + \frac{\gamma_2 k_2}{A_2} v_2 \\ \frac{dh_3}{dt} &= -\frac{a_3}{A_3} \sqrt{2gh_3} + \frac{(1-\gamma_2)k_2}{A_3} v_2 \\ \frac{dh_4}{dt} &= -\frac{a_4}{A_4} \sqrt{2gh_4} + \frac{(1-\gamma_1)k_1}{A_4} v_1\end{aligned}$$

Control Objective

Both the levels of tank1 and tank2 should be controlled by manipulating voltages to the pumps.

Data for simulation:

$$A_1, A_3 = 28 \text{ cm}^2$$

$$A_2, A_4 = 32 \text{ cm}^2$$

$$a_1, a_3 = 0.071 \text{ cm}^2$$

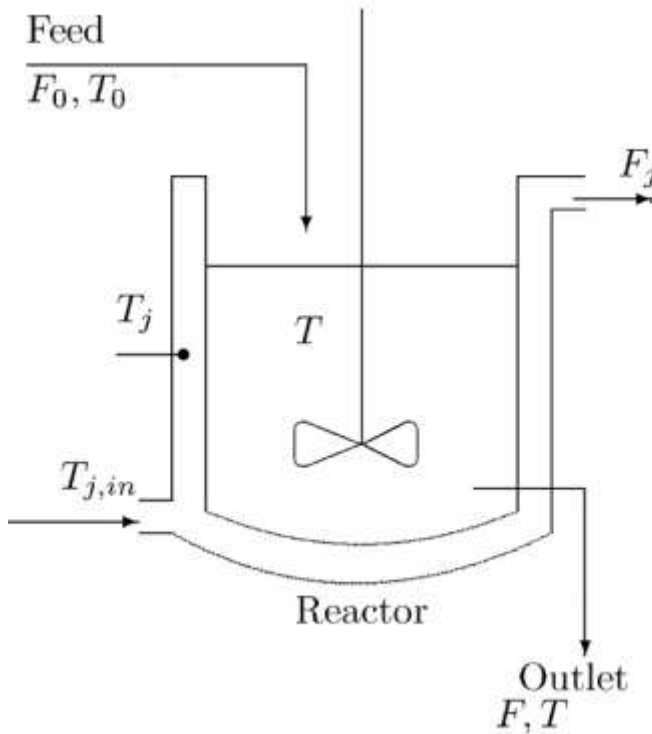
$$a_2, a_4 = 0.057 \text{ cm}^2$$

$$k_1, k_2 = 3.33, 3.35$$

$$v_1, v_2 = 3.0, 3.0$$

$$\gamma_1, \gamma_2 = 0.7, 0.6$$

Jacketed heated stirred tank



Assumptions:

Constant hold-up in tank and jacket

Constant heat capacities and densities

Incompressible flow

$$\frac{dV}{dt} = F_0 - F = 0$$

$$\frac{dT}{dt} = \frac{F}{V} (T_0 - T) + \frac{UA}{\rho C_p V} (T_j - T)$$

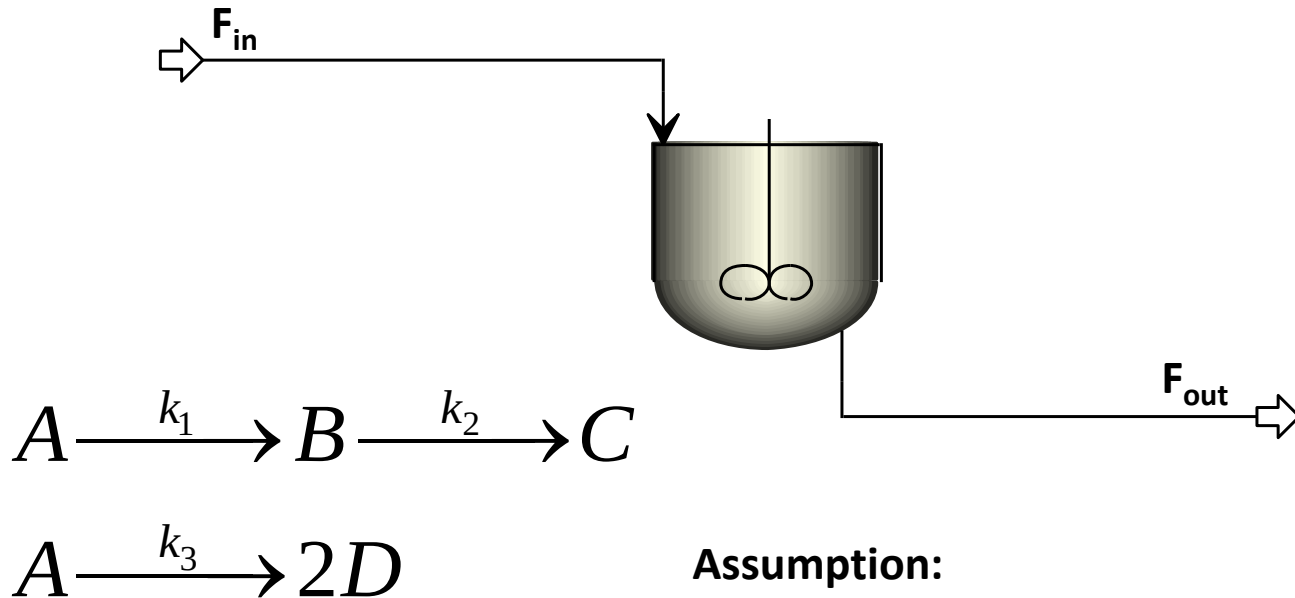
$$\frac{dT_j}{dt} = \frac{F_j}{V_j} (T_{ji} - T_j) - \frac{UA}{\rho_j C_{pj} V_j} (T_j - T)$$

Parameter and Steady State Values

$F = 30$ l/min; $F_j^s = 50$ l/min; $T_0 = 15^\circ\text{C}$; $T_{jin} = 93^\circ\text{C}$; $V = 300$ l; $V_j = 30$ l;

$\rho C_p = 1$ Kcal/K l; $\rho C_{pj} = 1.384$; $UA = 100$ Kcal/min K; $T^s = 60^\circ\text{C}$; $T_j = 73.5^\circ\text{C}$

Continuous Stirred Tank Reactor



Assumption:

1. Exit Condition = Reactor Condition
2. Isothermal Reaction
3. Constant volume, i.e, $F_{in} = F_{out} = F$
4. Only reactant A in feed is consumed, i.e, other reactant is in large excess. No product in the feed

Continuous Stirred Tank Reactor

Constant volume : $dV/dt = 0$

Dynamic model

- $\frac{dC_A}{dt} = \frac{F}{V} (C_{Af} - C_A) - k_1 C_A - k_3 C_A^2$
- $\frac{dC_B}{dt} = -\frac{F}{V} C_B + k_1 C_A - k_2 C_B$
- $\frac{dC_C}{dt} = -\frac{F}{V} C_C + k_2 C_B$
- $\frac{dC_D}{dt} = -\frac{F}{V} C_D + \frac{1}{2} k_3 C_A^2$

Data for the CSTR

$$V = 1 \text{ l}$$

$$F = 78 \text{ l/h}$$

$$C_{Af} = 10 \text{ mol/l}$$

$$k_1 = 50 \text{ h}^{-1}$$

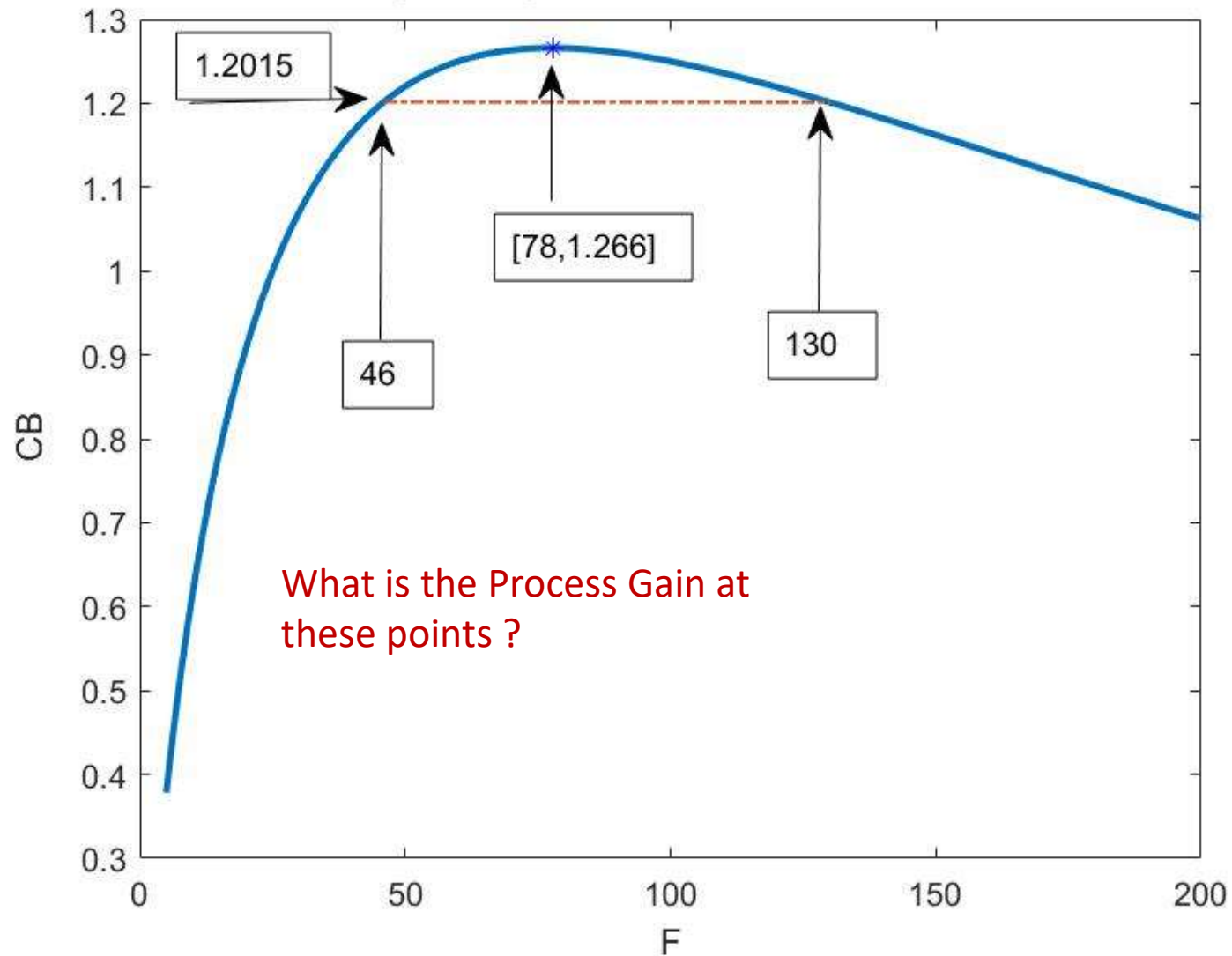
$$k_2 = 100 \text{ h}^{-1}$$

$$k_3 = 10 \text{ l.mol}^{-1}.\text{h}^{-1}$$

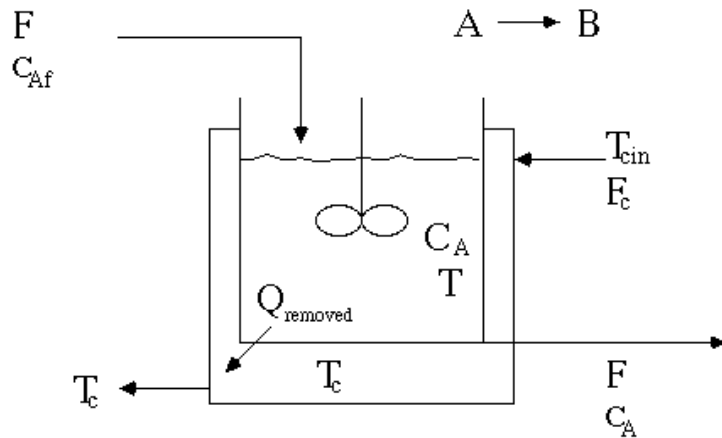
Since our objective is to control C_B , the reduced order model should be

- $\frac{dC_A}{dt} = \frac{F}{V} (C_{Af} - C_A) - k_1 C_A - k_3 C_A^2 = f_1(C_A, C_B, F)$
- $\frac{dC_B}{dt} = -\frac{F}{V} C_B + k_1 C_A - k_2 C_B = f_2(C_A, C_B, F)$

Steady State plot of Van-De-Vusse Reactor



Non-Isothermal Jacketed CSTR



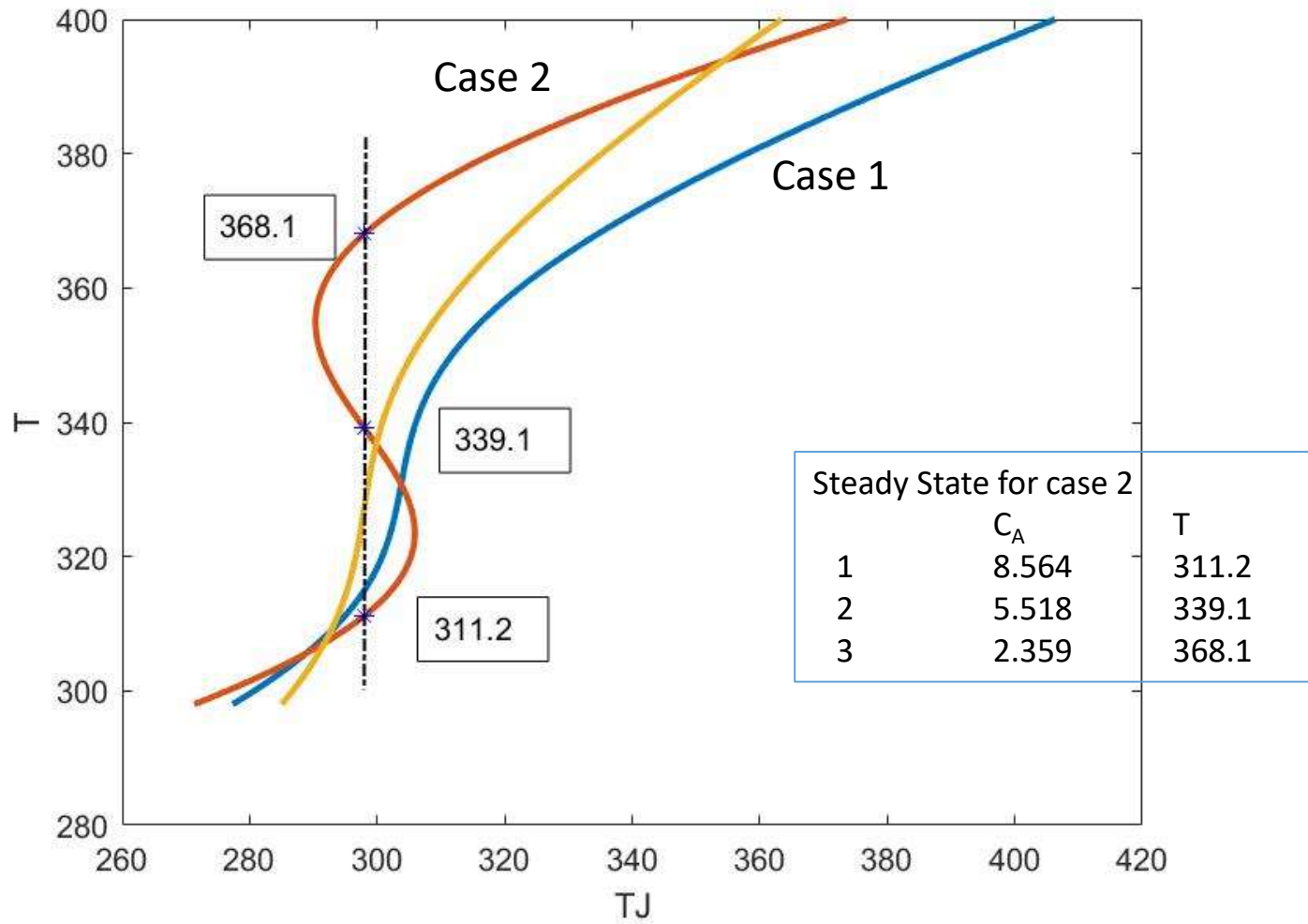
$$\frac{dC_A}{dt} = \frac{F}{V} (C_{Af} - C_A) - r$$

$$\frac{dT}{dt} = \frac{F}{V} (T_f - T) + \left(\frac{-\Delta H}{\rho C_p} \right) r - \frac{UA}{V\rho C_p} (T - T_c)$$

$$r = k_0 e^{-\frac{E}{RT}} C_A$$

parameter	case 1	case 2	case 3
F/V, hr-1	1	1	1
k_0 , hr-1	14,825*3600	9,703*3600	18,194*3600
$(-\Delta H)$, kcal/kgmol	5215	5960	8195
E, kcal/kgmol	11,843	11,843	11,843
ρc_p , kcal/(m ³ °C)	500	500	500
T_f , °C	25	25	25
C_{Af} , kgmol/m ³	10	10	10
UA/V, kcal/(m ³ °C hr)	250	150	750
T_c , °C	25	25	25

Variation of Reactor Temp with Jacket Temperature



Styrene Polymerization Reactor

x_1 Initiator Conc
 x_2 Monomer Conc
 x_3 Reactor Temp
 x_4 Jacket Temp.

$$\begin{aligned}
 \dot{x}_1 &= \frac{(F_i C_{ia} - F_o x_1)}{V} - k_d x_1 \\
 \dot{x}_2 &= \frac{(F_m C_{ma} - F_o x_2)}{V} - k_p x_2 \mathcal{R} \\
 \dot{x}_3 &= \frac{F_o (T_a - x_3)}{V} - \frac{\Delta H}{\rho C_p} k_p x_2 \mathcal{R} - \frac{UA}{\rho C_p V} (x_3 - x_4) \\
 \dot{x}_4 &= \frac{F_j (T_{j,in} - x_4)}{V_j} + \frac{UA}{\rho_j C_{pj} V_j} (x_3 - x_4)
 \end{aligned}$$

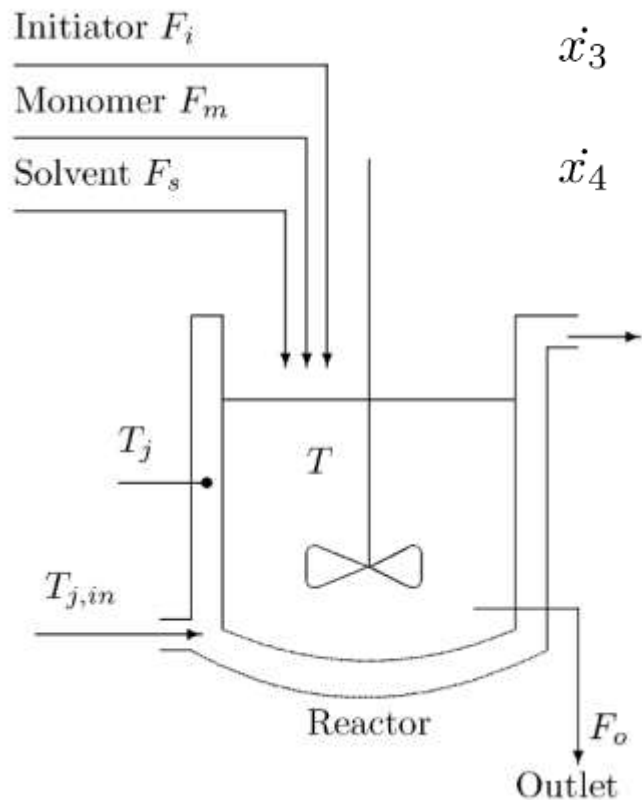
$$F_o = F_i + F_m + F_s.$$

The chain concentration of growing polymer is equal to

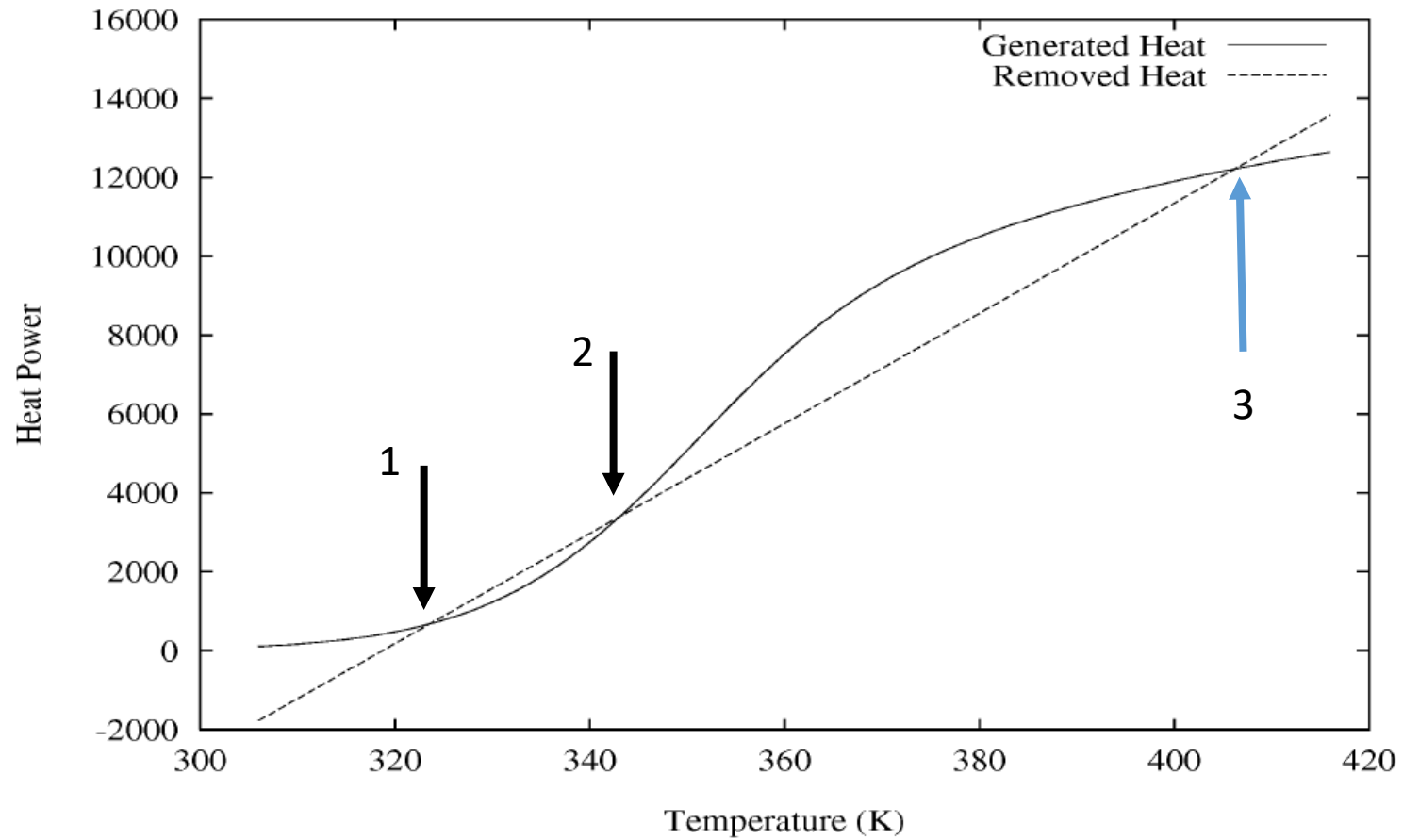
$$\mathcal{R} = (2 f k_d x_1 / k_t)^{0.5}$$

The dissociation, propagation and termination
Rate constants follow Arrhenius law :

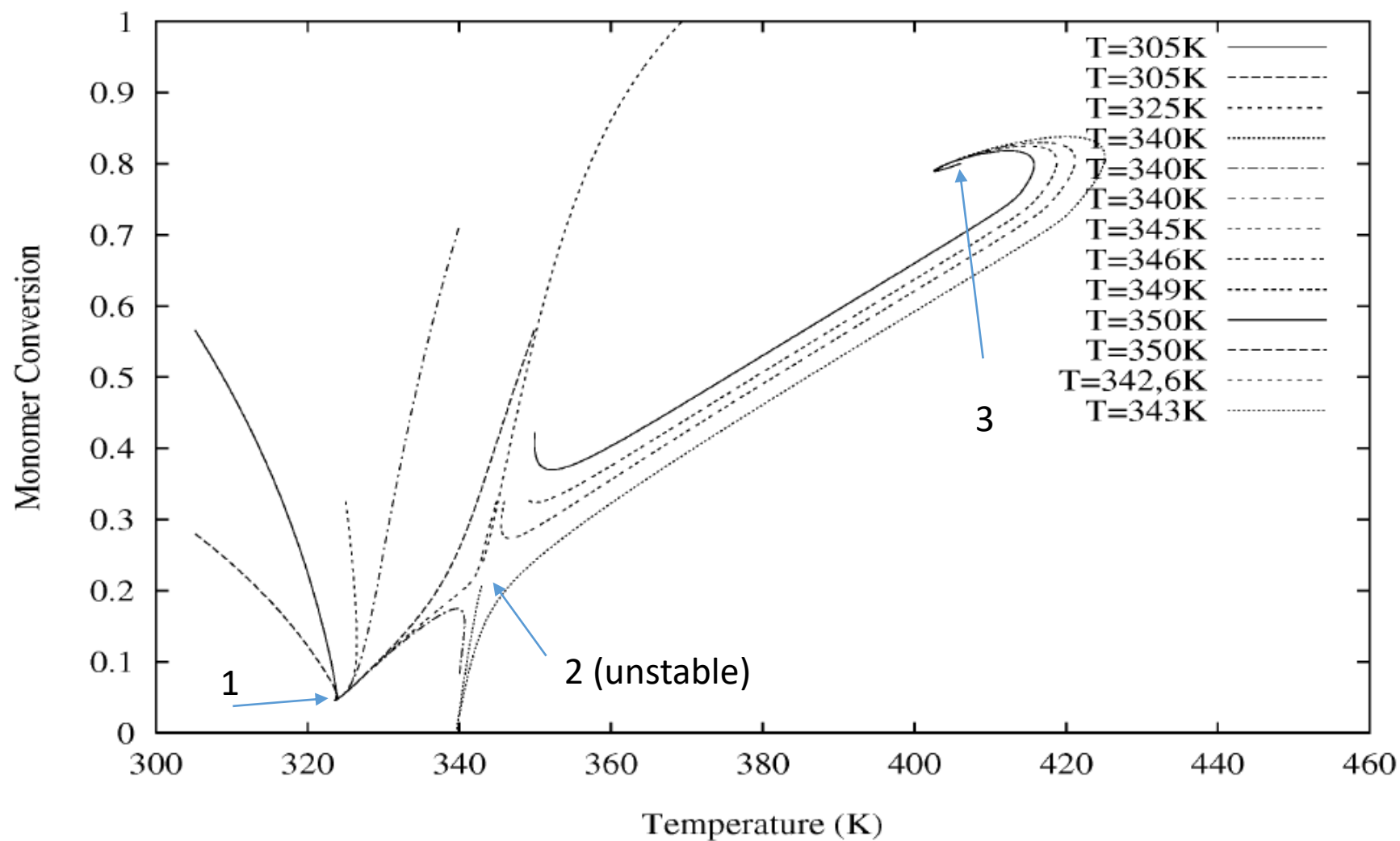
$$k_i = k_{i0} e^{-\frac{E_i}{RT}} \quad \text{for } i = d, p \text{ or } t$$



Steady State Heat Power vs Reactor Temperature



Phase Portrait



Process Dynamics & Control

Development of Mathematical Model

Motivation:

- Develop understanding of process
- Build a mathematical hypothesis of process mechanisms
- Match observed process behavior
- useful in design, optimization and *control* of process

Development of Mathematical Model

Motivation:

- Develop understanding of process
- Build a mathematical hypothesis of process mechanisms
- Match observed process behavior
- useful in design, optimization and *control* of process

Control:

- Interested in description of process dynamics
 - Dynamic model is used to predict how process responds to given input
 - Tells us how to react

Development of Mathematical Model

Mathematical Modeling of a process require

- Understanding of the process
- Writing conservation balance for Mass, Momentum and Energy
- Writing closure equations, i.e, constitutive relationships defining equilibrium and rate parameters
- Devising solution technique

What kind of model do we need?

Dynamic vs. Steady-state

- Steady-state
 - Variables not a function of time
 - useful for design calculation

What kind of model do we need?

Dynamic vs. Steady-state

- Steady-state
 - Variables not a function of time
 - useful for design calculation
- *Dynamic*
 - Variables are a function of time
 - Control requires dynamic model

What kind of model do we need?

Experimental vs Theoretical

- Experimental
 - Derived from tests performed on actual process
 - Simpler model forms
 - Easier to manipulate

What kind of model do we need?

Experimental vs Theoretical

- Experimental
 - Derived from tests performed on actual process
 - Simpler model forms
 - Easier to manipulate
- Theoretical
 - Application of fundamental laws of physics and chemistry
 - more complex but provides understanding
 - Required in design stages

What kind of model do we need?

Empirical vs. Mechanistic models

- Empirical Models
 - only local representation of the process (no extrapolation)
 - model only as good as the data
 - do not rely on underlying mechanisms
 - Fit specific function to match process

What kind of model do we need?

Empirical vs. Mechanistic models

- Empirical Models
 - only local representation of the process (no extrapolation)
 - model only as good as the data
 - do not rely on underlying mechanisms
 - Fit specific function to match process
- Mechanistic Models
 - Rely on our understanding of a process
 - Derived from first principles
 - Observing laws of conservation of
 - Mass
 - Energy
 - Momentum
 - Useful for simulation and exploration of new operating conditions
 - May contain unknown constants that must be estimated

What kind of model do we need?

Linear vs Nonlinear

- Linear
 - basis for most industrial control
 - simpler model form, easy to identify
 - easy to design controller
 - poor prediction, adequate control

What kind of model do we need?

Linear vs Nonlinear

- Linear
 - basis for most industrial control
 - simpler model form, easy to identify
 - easy to design controller
 - poor prediction, adequate control
- Nonlinear
 - Reality
 - more complex and difficult to identify
 - need state-of-the-art controller design techniques to do the job
 - better prediction and control

What kind of model do we need?

In existing processes, we rely on

- Dynamic models obtained from experiments
- Usually of an empirical nature
- Linear

In new applications (or difficult problems)

- Focus on mechanistic modeling
- Dynamic models derived from theory
- Nonlinear

General modeling procedure

1. Identify modeling objectives
 - end use of model (e.g. control)
2. Identify fundamental quantities of interest
 - Mass, Energy and/or Momentum
3. Identify boundaries
4. Apply fundamental physical and chemical laws
 - Mass, Energy and/or Momentum balances
5. Make appropriate assumptions (Simplify)
 - ideality (e.g. isothermal, adiabatic, ideal gas, no friction, incompressible flow, etc,...)
6. Write down energy, mass and momentum balances
 - develop the model equations

General modeling procedure

7. Check model consistency

- do we have more unknowns than equations

8. Determine unknown constants

- e.g. friction coefficients, fluid density and viscosity

9. Solve model equations

- typically nonlinear ordinary (or partial) differential equations
- initial value problems

10. Check the validity of the model

- compare to process behavior

Benchmark Systems to be used for the course

- Tank Level System
 - 3 tank in series
 - Quadruple Tank
- Continuous Stirred Tank Heater
- Continuous Stirred Tank Reactor system
 - Isothermal Reactor (Van de Vusse reaction)
 - Non-isothermal Reactor
 - Polymerization Reactor
- Distillation Column

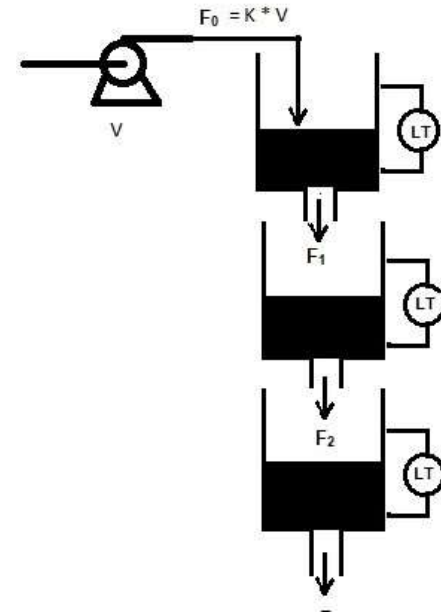
MATLAB / SIMULINK model will be developed based on nonlinear model

Mathematical Model of tank level system

$$\frac{dh_1}{dt} = \frac{F_0}{A_1} - \frac{a_1}{A_1} \sqrt{2gh_1}$$

$$\frac{dh_2}{dt} = \frac{a_1}{A_2} \sqrt{2gh_1} - \frac{a_2}{A_2} \sqrt{2gh_2}$$

$$\frac{dh_3}{dt} = \frac{a_2}{A_3} \sqrt{2gh_2} - \frac{a_3}{A_3} \sqrt{2gh_3}$$



Control objective:

Control the 3rd tank level by manipulating pump voltage V , where $F_0 = K V$

Add q_2 input flow on the tanks 2 and set q_2 as manipulated variable.

Which one is difficult to control?

Data for Simulation in Matlab/Simulink

$$A_1, A_3 = 28 \text{ cm}^2 \quad A_2 = 32 \text{ cm}^2 \quad V^s = 3 \text{ V}$$

$$a_1 = 0.06725 \text{ cm}^2 \quad a_2 = 0.05683 \text{ cm}^2$$

$$a_3 = 0.07089 \text{ cm}^2 \quad K=3.14$$

- Transfer Function model

Define : $x_1 = h_1 - h_1^s$; $x_2 = h_2 - h_2^s$; $x_3 = h_3 - h_3^s$; $u = V - V^s$

Linearizing around steady state value h_1^s

$$\frac{dh_1}{dt} = \frac{K}{A_1} V - \frac{a_1}{A_1} \sqrt{2g} \left[\sqrt{h_1^s} + \frac{1}{2\sqrt{h_1^s}} (h_1 - h_1^s) \right]$$

Subtracting above from steady state equation,

$$\frac{dx_1}{dt} = \frac{K}{A_1} u - \frac{a_1 \sqrt{g}}{A_1 \sqrt{2h_1^s}} x_1 \quad \Rightarrow \text{Transfer function } \frac{x_1(s)}{u(s)} = \frac{K_{p1}}{\tau_{p1}s+1}$$

$$\text{Where, } K_{p1} = \frac{K}{a_1} \sqrt{\frac{2h_1^s}{g}} \quad \text{and } \tau_{p1} = \frac{A_1}{a_1} \sqrt{\frac{2h_1^s}{g}}$$

Similarly,

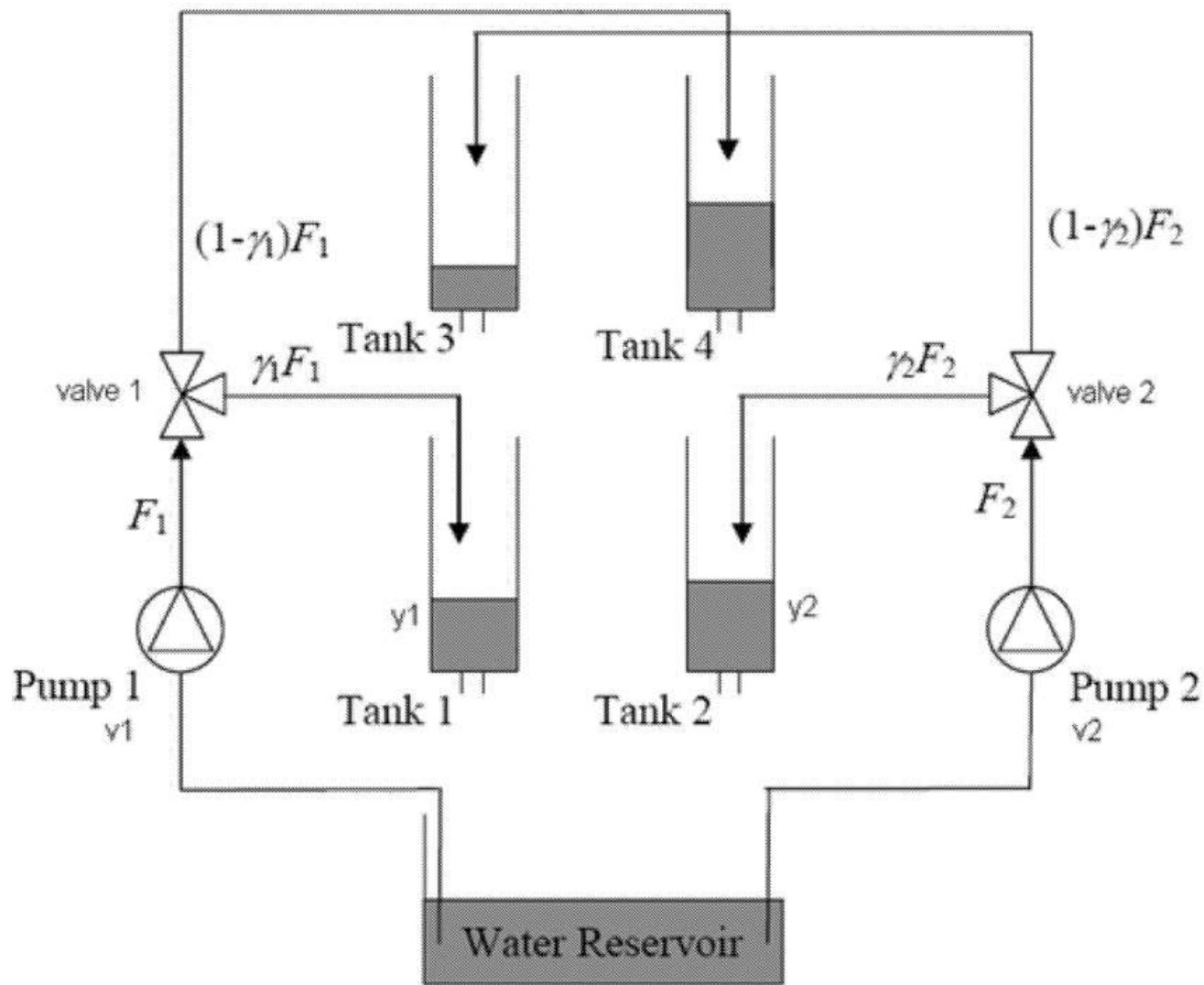
$$\frac{dx_2}{dt} = \frac{a_1}{A_2} \sqrt{\frac{g}{2h_1^s}} x_1 - \frac{a_2}{A_2} \sqrt{\frac{g}{2h_2^s}} x_2 \quad \text{and Transfer function } \frac{x_2(s)}{x_1(s)} = \frac{K_{p2}}{\tau_{p2}s+1}$$

$$\frac{dx_3}{dt} = \frac{a_2}{A_3} \sqrt{\frac{g}{2h_2^s}} x_2 - \frac{a_3}{A_3} \sqrt{\frac{g}{2h_3^s}} x_3 \quad \text{and Transfer function } \frac{x_3(s)}{x_2(s)} = \frac{K_{p3}}{\tau_{p3}s+1}$$

$$\text{Where, } K_{pj} = \frac{a_{j-1}}{a_j} \sqrt{\frac{h_j^s}{h_{j-1}^s}} \quad \text{and } \tau_{pj} = \frac{A_j}{a_j} \sqrt{\frac{2h_j^s}{g}} \quad \text{for } j = 2, 3$$

$$\text{So, Process Transfer function } G(s) = \frac{x_3(s)}{u(s)} = \frac{K_{p1} K_{p2} K_{p3}}{(\tau_{p1}s+1)(\tau_{p2}s+1)(\tau_{p3}s+1)}$$

Quadruple Tank problem



Quadruple Tank problem

$$\begin{aligned}\frac{dh_1}{dt} &= -\frac{a_1}{A_1} \sqrt{2gh_1} + \frac{a_3}{A_1} \sqrt{2gh_3} + \frac{\gamma_1 k_1}{A_1} v_1 \\ \frac{dh_2}{dt} &= -\frac{a_2}{A_2} \sqrt{2gh_2} + \frac{a_4}{A_2} \sqrt{2gh_4} + \frac{\gamma_2 k_2}{A_2} v_2 \\ \frac{dh_3}{dt} &= -\frac{a_3}{A_3} \sqrt{2gh_3} + \frac{(1-\gamma_2)k_2}{A_3} v_2 \\ \frac{dh_4}{dt} &= -\frac{a_4}{A_4} \sqrt{2gh_4} + \frac{(1-\gamma_1)k_1}{A_4} v_1\end{aligned}$$

Control Objective

Both the levels of tank1 and tank2 should be controlled by manipulating voltages to the pumps.

Data for simulation:

$$A_1, A_3 = 28 \text{ cm}^2$$

$$A_2, A_4 = 32 \text{ cm}^2$$

$$a_1, a_3 = 0.071 \text{ cm}^2$$

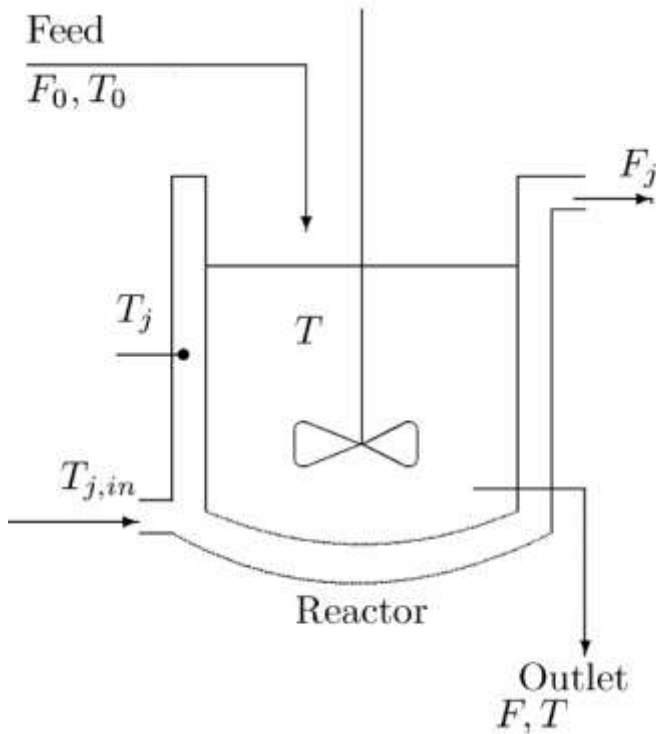
$$a_2, a_4 = 0.057 \text{ cm}^2$$

$$k_1, k_2 = 3.33, 3.35$$

$$v_1, v_2 = 3.0, 3.0$$

$$\gamma_1, \gamma_2 = 0.7, 0.6$$

Jacketed heated stirred tank



Assumptions:

Constant hold-up in tank and jacket

Constant heat capacities and densities

Incompressible flow

$$\frac{dV}{dt} = F_0 - F = 0$$

$$\frac{dT}{dt} = \frac{F}{V} (T_0 - T) + \frac{UA}{\rho C_p V} (T_j - T)$$

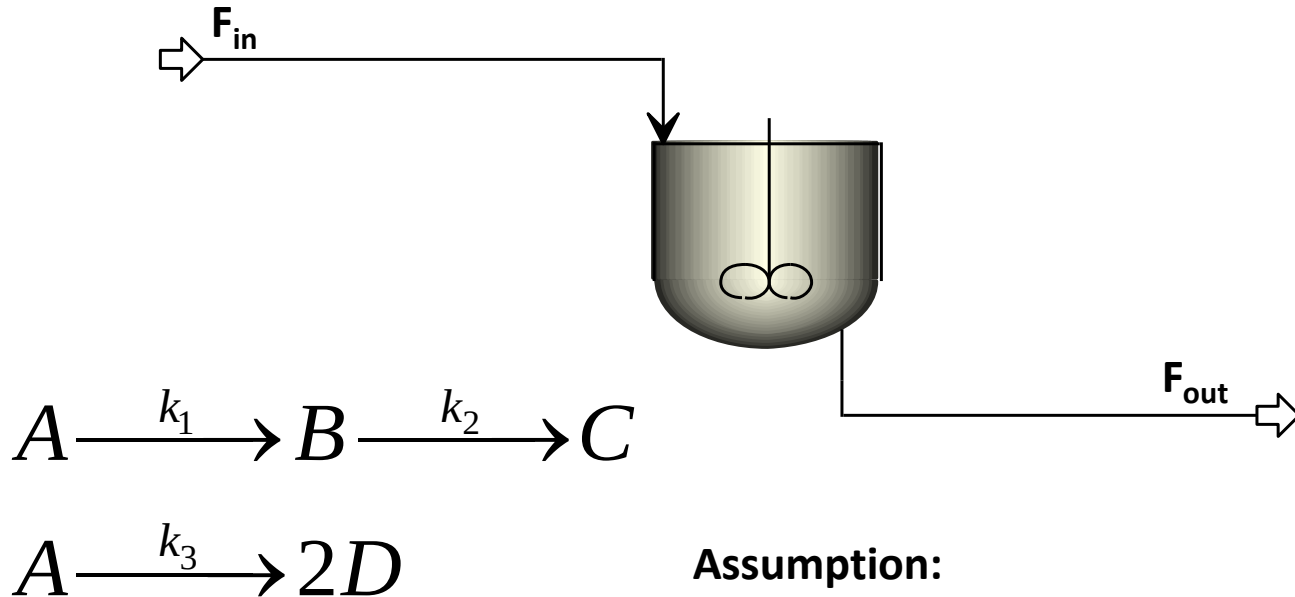
$$\frac{dT_j}{dt} = \frac{F_j}{V_j} (T_{j,i} - T_j) - \frac{UA}{\rho_j C_{pj} V_j} (T_j - T)$$

Parameter and Steady State Values

$F = 30$ l/min; $F_j^s = 50$ l/min; $T_0 = 15^\circ\text{C}$; $T_{j,i} = 93^\circ\text{C}$; $V = 300$ l; $V_j = 30$ l;

$\rho C_p = 1$ Kcal/K l; $\rho C_{pj} = 1.384$; $UA = 100$ Kcal/min K; $T^s = 60^\circ\text{C}$; $T_j = 73.5^\circ\text{C}$

Continuous Stirred Tank Reactor



Assumption:

1. Exit Condition = Reactor Condition
2. Isothermal Reaction
3. Constant volume, i.e, $F_{in} = F_{out} = F$
4. Only reactant A in feed is consumed, i.e, other reactant is in large excess. No product in the feed

Continuous Stirred Tank Reactor

Constant volume : $dV/dt = 0$

Dynamic model

- $\frac{dC_A}{dt} = \frac{F}{V} (C_{Af} - C_A) - k_1 C_A - k_3 C_A^2$
- $\frac{dC_B}{dt} = -\frac{F}{V} C_B + k_1 C_A - k_2 C_B$
- $\frac{dC_C}{dt} = -\frac{F}{V} C_C + k_2 C_B$
- $\frac{dC_D}{dt} = -\frac{F}{V} C_D + \frac{1}{2} k_3 C_A^2$

Data for the CSTR

$$V = 1 \text{ l}$$

$$F = 78 \text{ l/h}$$

$$C_{Af} = 10 \text{ mol/l}$$

$$k_1 = 50 \text{ h}^{-1}$$

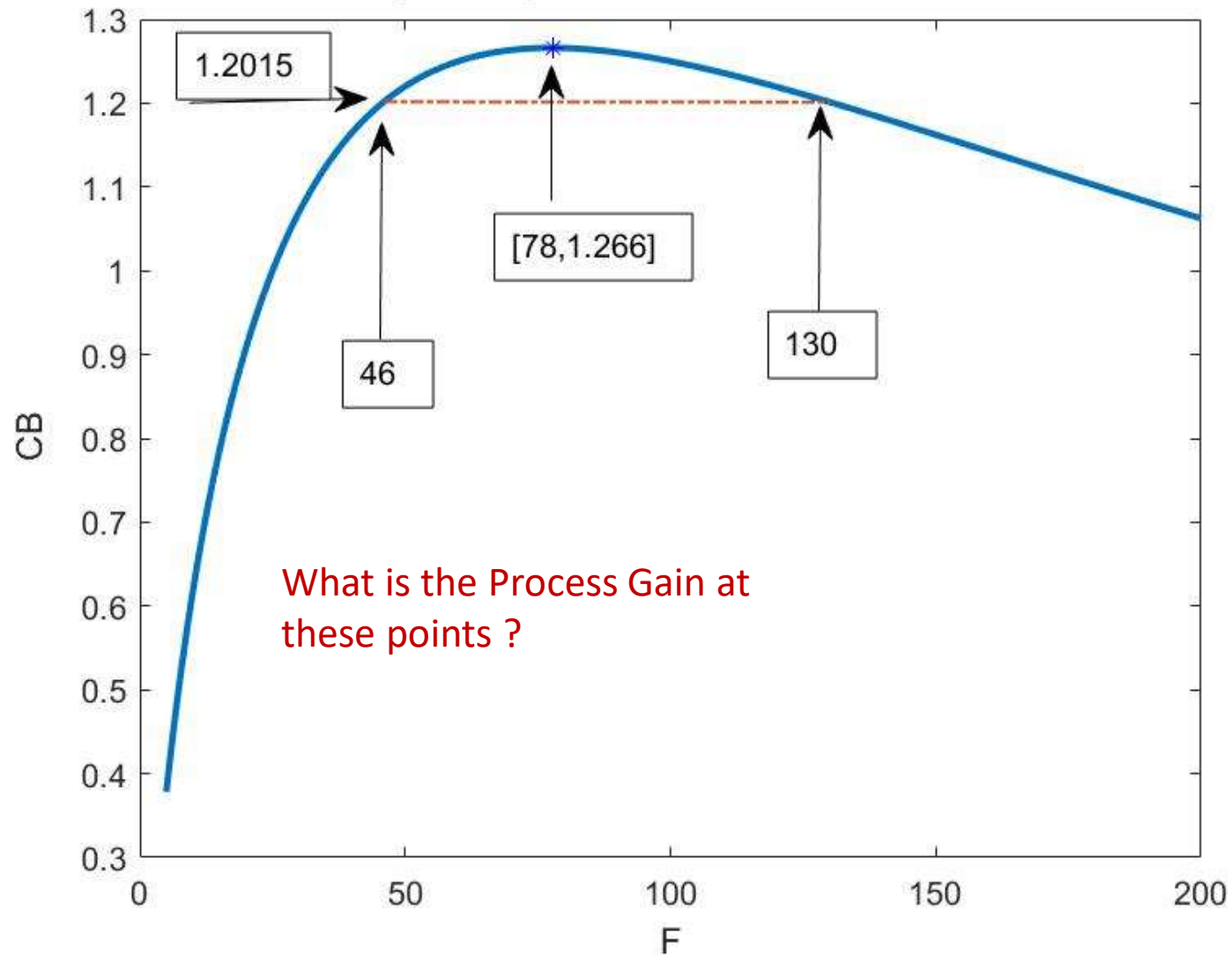
$$k_2 = 100 \text{ h}^{-1}$$

$$k_3 = 10 \text{ l.mol}^{-1}.\text{h}^{-1}$$

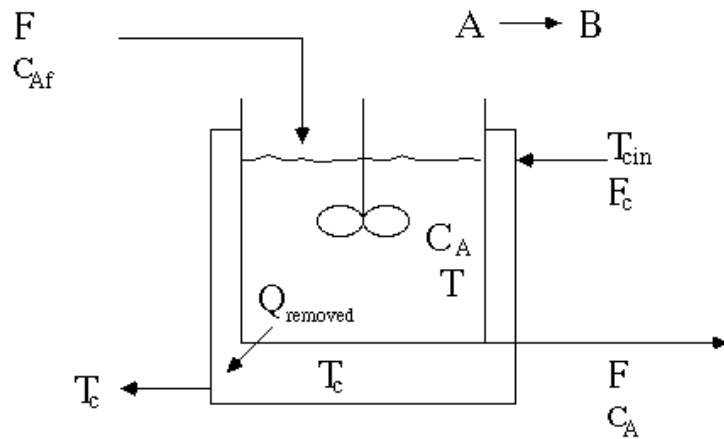
Since our objective is to control C_B , the reduced order model should be

- $\frac{dC_A}{dt} = \frac{F}{V} (C_{Af} - C_A) - k_1 C_A - k_3 C_A^2 = f_1(C_A, C_B, F)$
- $\frac{dC_B}{dt} = -\frac{F}{V} C_B + k_1 C_A - k_2 C_B = f_2(C_A, C_B, F)$

Steady State plot of Van-De-Vusse Reactor



Non-Isothermal Jacketed CSTR



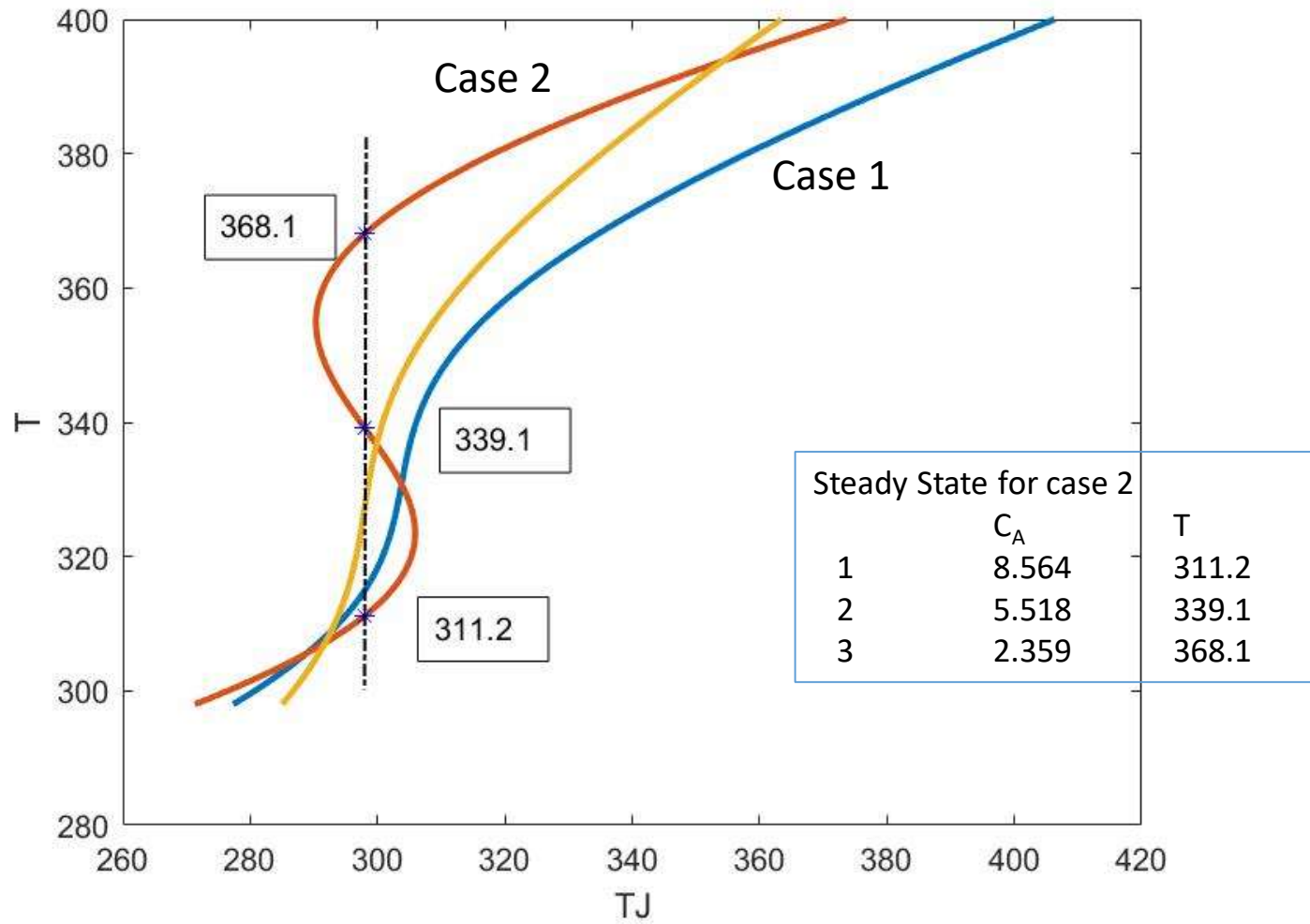
$$\frac{dC_A}{dt} = \frac{F}{V} (C_{Af} - C_A) - r$$

$$\frac{dT}{dt} = \frac{F}{V} (T_f - T) + \left(\frac{-\Delta H}{\rho C_p} \right) r - \frac{UA}{V\rho C_p} (T - T_c)$$

$$r = k_0 e^{-\frac{E}{RT}} C_A$$

parameter	case 1	case 2	case 3
F/V, hr-1	1	1	1
k ₀ , hr-1	14,825*3600	9,703*3600	18,194*3600
(-ΔH), kcal/kgmol	5215	5960	8195
E, kcal/kgmol	11,843	11,843	11,843
ρc _p , kcal/(m ³ °C)	500	500	500
T _f , °C	25	25	25
C _{Af} , kgmol/m ³	10	10	10
UA/V, kcal/(m ³ °C hr)	250	150	750
T _c , °C	25	25	25

Variation of Reactor Temp with Jacket Temperature



Styrene Polymerization Reactor

x_1 Initiator Conc
 x_2 Monomer Conc
 x_3 Reactor Temp
 x_4 Jacket Temp.

$$\begin{aligned}
 \dot{x}_1 &= \frac{(F_i C_{ia} - F_o x_1)}{V} - k_d x_1 \\
 \dot{x}_2 &= \frac{(F_m C_{ma} - F_o x_2)}{V} - k_p x_2 \mathcal{R} \\
 \dot{x}_3 &= \frac{F_o (T_a - x_3)}{V} - \frac{\Delta H}{\rho C_p} k_p x_2 \mathcal{R} - \frac{UA}{\rho C_p V} (x_3 - x_4) \\
 \dot{x}_4 &= \frac{F_j (T_{j,in} - x_4)}{V_j} + \frac{UA}{\rho_j C_{pj} V_j} (x_3 - x_4)
 \end{aligned}$$

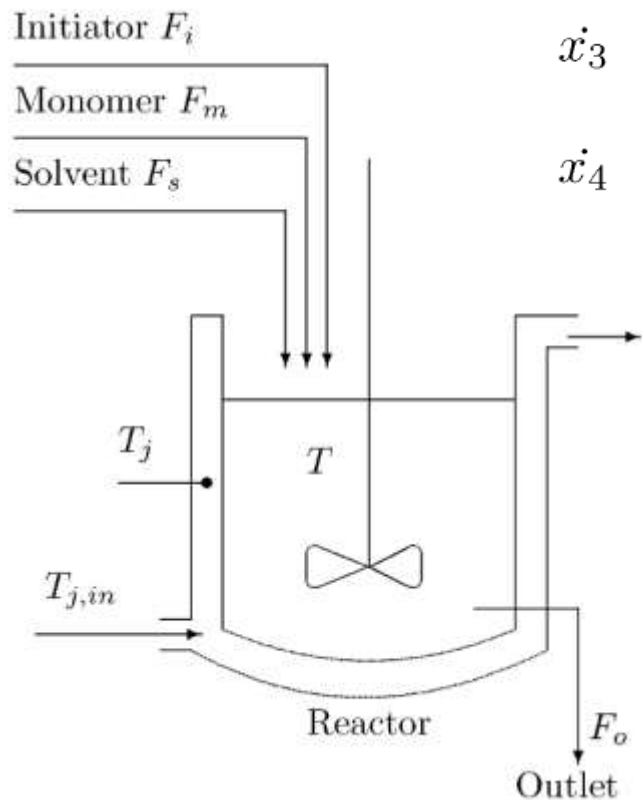
$$F_o = F_i + F_m + F_s.$$

The chain concentration of growing polymer is equal to

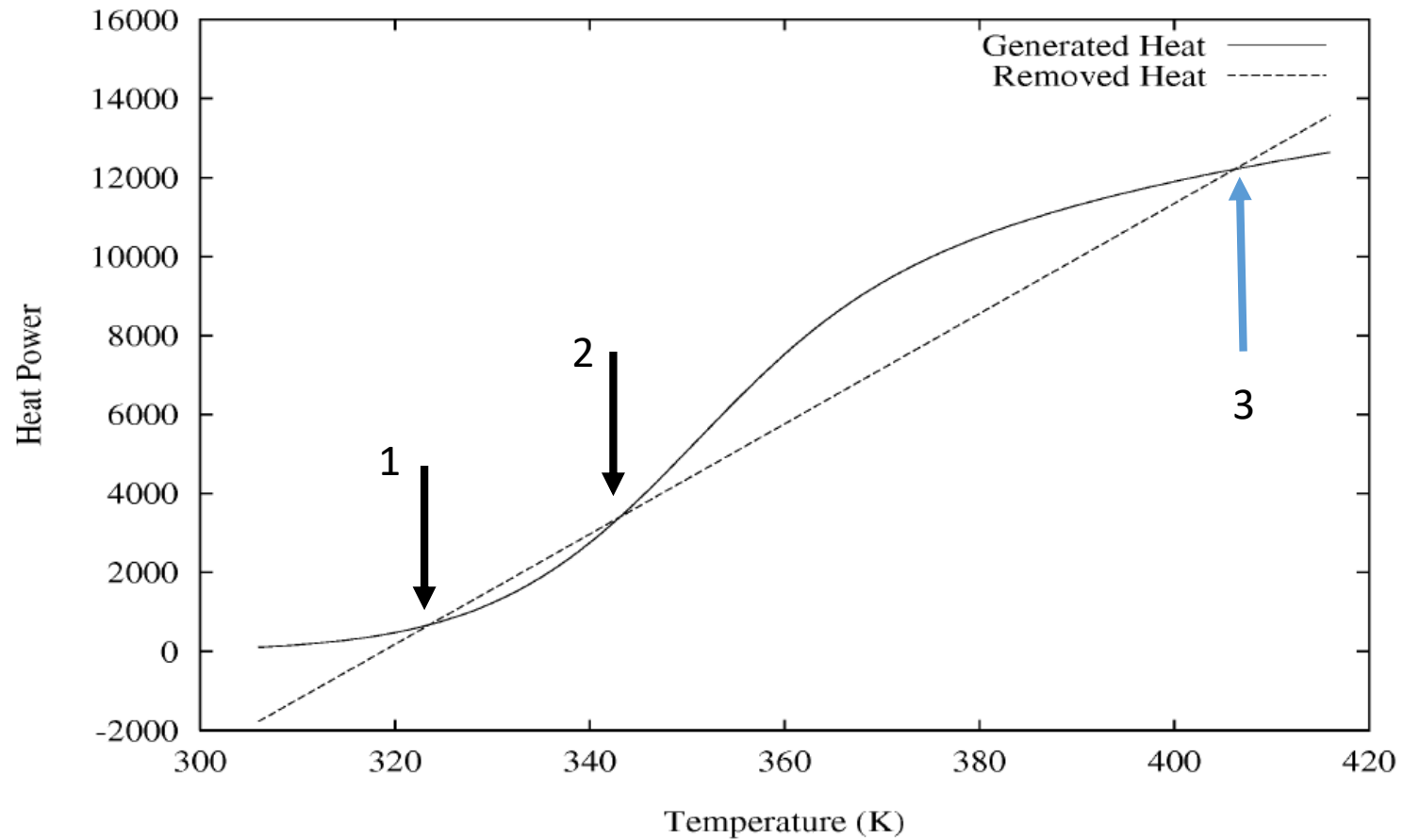
$$\mathcal{R} = (2 f k_d x_1 / k_t)^{0.5}$$

The dissociation, propagation and termination
Rate constants follow Arrhenius law :

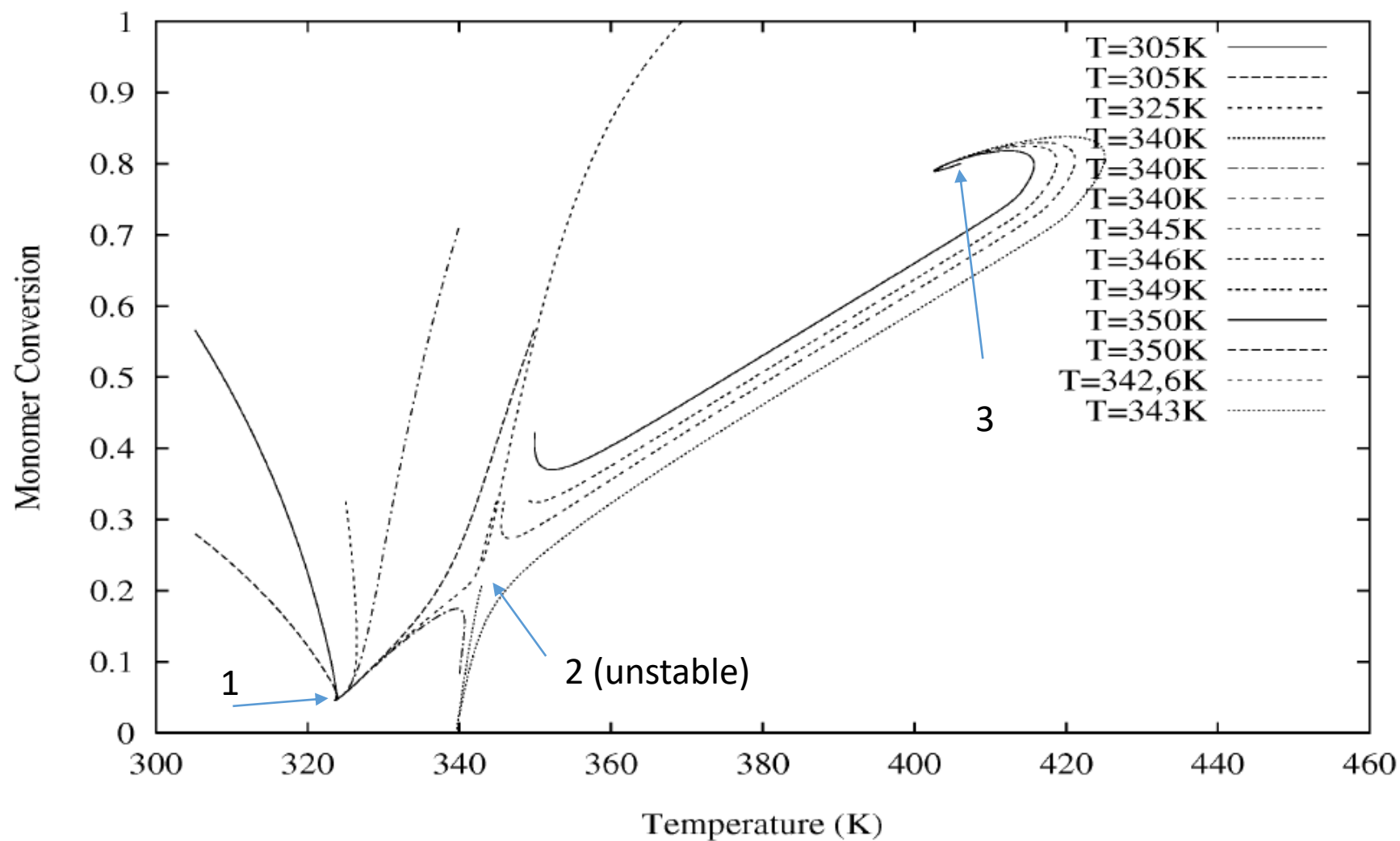
$$k_i = k_{i0} e^{-\frac{E_i}{RT}} \quad \text{for } i = d, p \text{ or } t$$



Steady State Heat Power vs Reactor Temperature

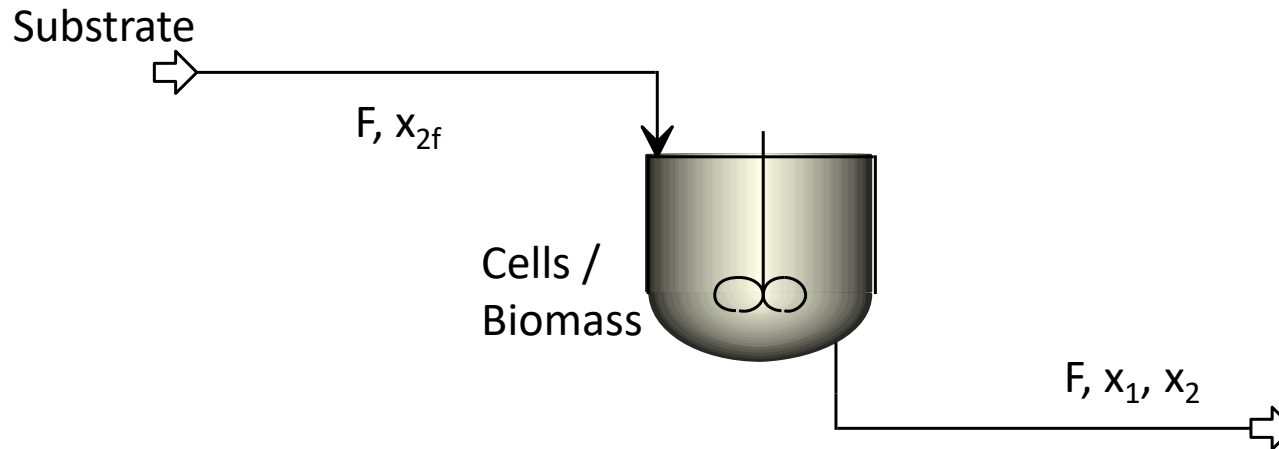


Phase Portrait



Process Dynamics & Control

Biochemical Reactor



x_1 is mass of cells per unit volume in the reactor

x_2 is mass of substrate per unit volume in the reactor

x_{2f} is mass of substrate per unit volume in the feed stream

Assumption:

1. Exit Condition = Reactor Condition
2. Isothermal Reaction
3. Constant volume, i.e, $F_{in} = F_{out} = F$
4. No cell/biomass is present in the feed stream.

Biochemical Reactor (Dynamic Model)

Dynamic Model

$$\frac{dx_1}{dt} = -\frac{F}{V}x_1 + \mu x_1 = (\mu - D)x_1$$

$$\frac{dx_2}{dt} = D(x_{2f} - x_2) - \frac{\mu x_1}{Y}$$

Where yield, $Y = \frac{\text{mass of cells produced}}{\text{mass of substrate consumed}}$ and μ is specific growth rate coefficient for cell mass which is not constant but function of substrate concentration. The most common functions are:

Monod model:

$$\mu = \frac{\mu_{max} x_2}{k_m + x_2}$$

Substrate inhibition model:

$$\mu = \frac{\mu_{max} x_2}{k_m + x_2 + k_1 x_2^2}$$

Biochemical Reactor

Steady State equations:

$$\frac{dx_1}{dt} = 0 = -\frac{F}{V}x_1 + \mu x_1 = (\mu - D)x_1 = \left(\frac{\mu_{max} x_2}{k_m + x_2} - D\right)x_1$$

$$\text{if, } x_1 \neq 0, \text{ then } x_2 = \frac{DK_m}{\mu_{max} - D}$$

$$\frac{dx_2}{dt} = 0 = D(x_{2f} - x_2) - \frac{\mu x_1}{Y}$$

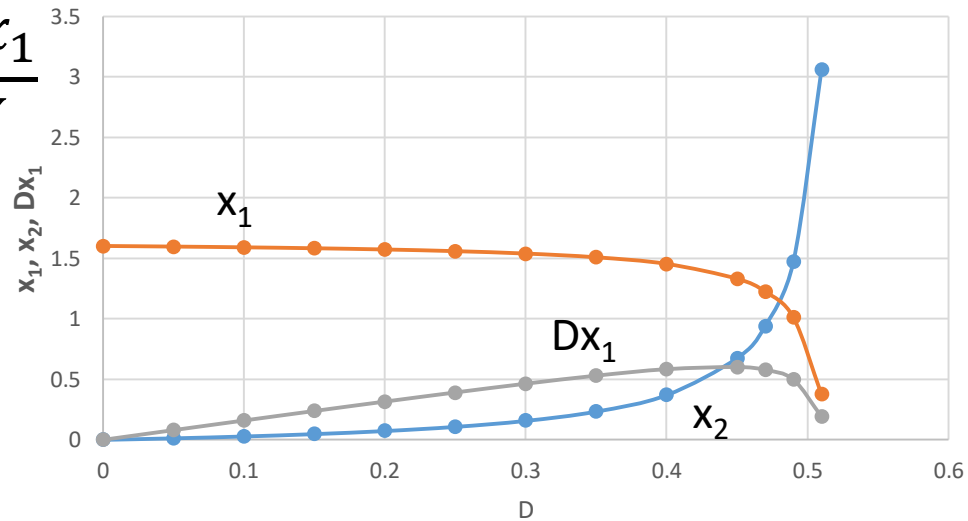
$$\text{So, } x_1 = Y\left(x_{2f} - \frac{DK_m}{\mu_{max} - D}\right)$$

Dx_1 is the rate of cell

Production per unit reactor

Volume.

Effect of Dilution Rate



Biochemical Reactor

We need to find optimum dilution rate to maximize Dx_1 . so,

$$Dx_1 = Y(Dx_{2f} - \frac{D^2 K_m}{\mu_{max} - D})$$

$$\frac{d(\frac{Dx_1}{Y})}{dD} = x_{2f} - \frac{(\mu_{max} - D)2DK_m - D^2 K_m(-1)}{(\mu_{max} - D)^2} = 0$$

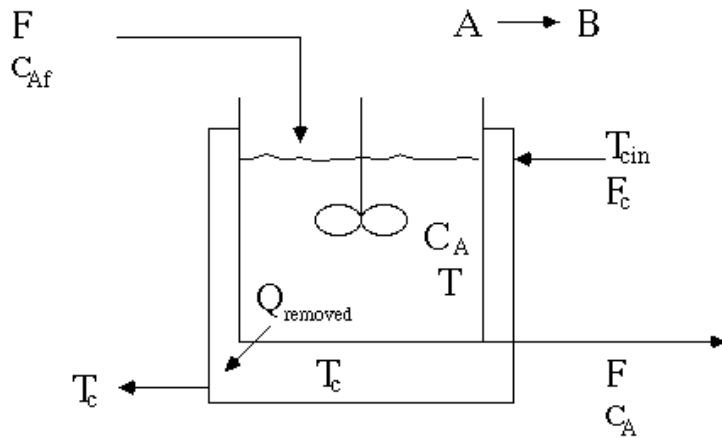
$$\text{Or, } D^2(x_{2f} + k_m) - 2D\mu_{max}(x_{2f} + k_m) + x_{2f}\mu_{max}^2 = 0$$

Adding $K_m\mu_{max}^2$ both sides, we get

$$(D - \mu_{max})^2 = \frac{K_m\mu_{max}^2}{x_{2f} + K_m}$$

$$\text{Or } D_{opt} = \mu_{max} \left(1 - \sqrt{\frac{K_m}{x_{2f} + K_m}} \right) \text{ since } D \text{ can not be greater than } \mu_{max}$$

Non-Isothermal Jacketed CSTR



$$\frac{dC_A}{dt} = \frac{F}{V} (C_{Af} - C_A) - r$$

$$\frac{dT}{dt} = \frac{F}{V} (T_f - T) + \left(\frac{-\Delta H}{\rho C_p} \right) r - \frac{UA}{V\rho C_p} (T - T_c)$$

$$\frac{dT_c}{dt} = \frac{F_c}{V_c} (T_{ci} - T_c) + \frac{UA}{V_c \rho_c C_{pc}} (T - T_c)$$

Data:

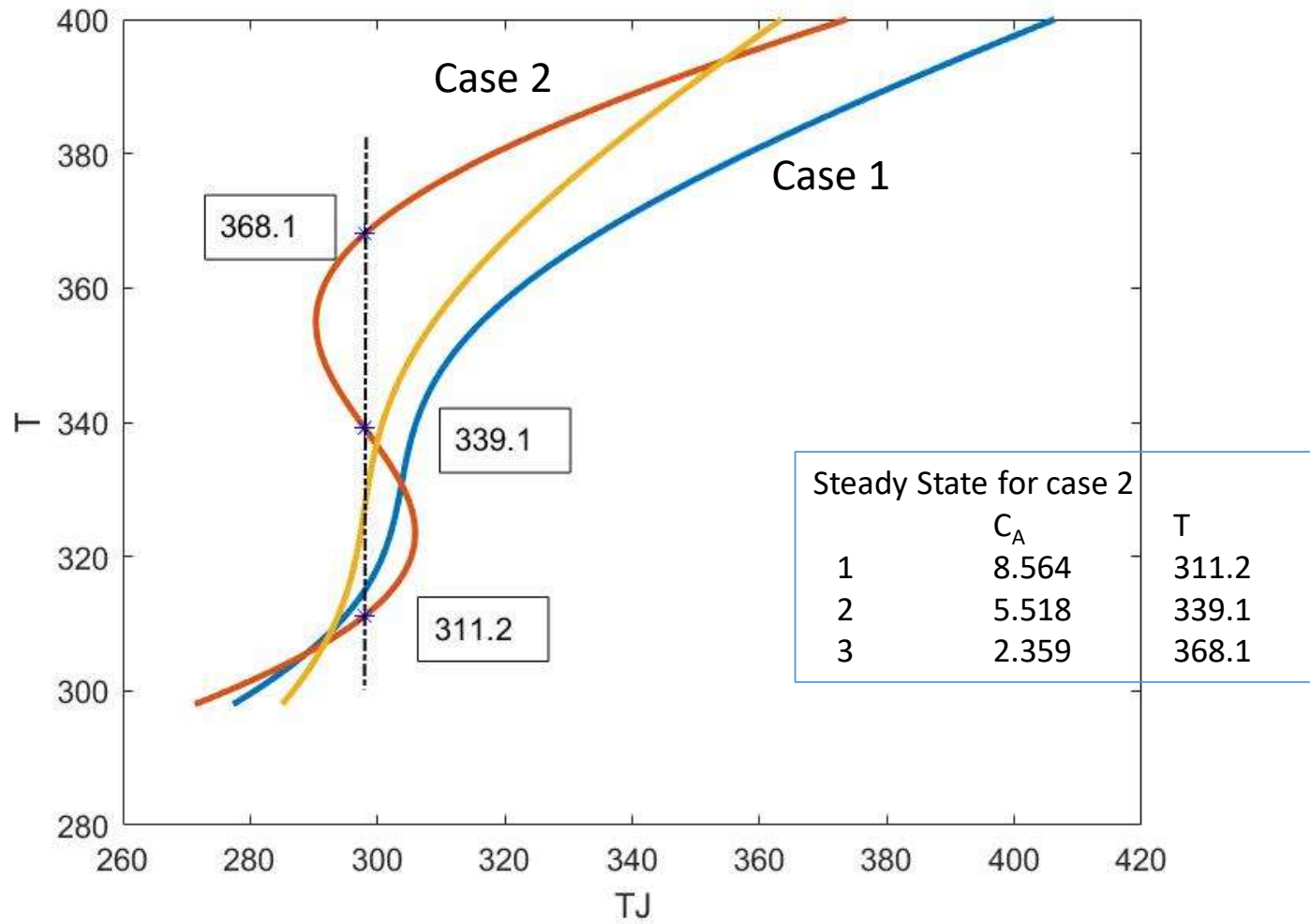
$F/V, \text{ hr}^{-1} = 1$; $E=11843 \text{ kcal/kgmol}$;

$\rho C_p=500$; $T_f = 25^\circ\text{C}$; $C_{af} = 10$,

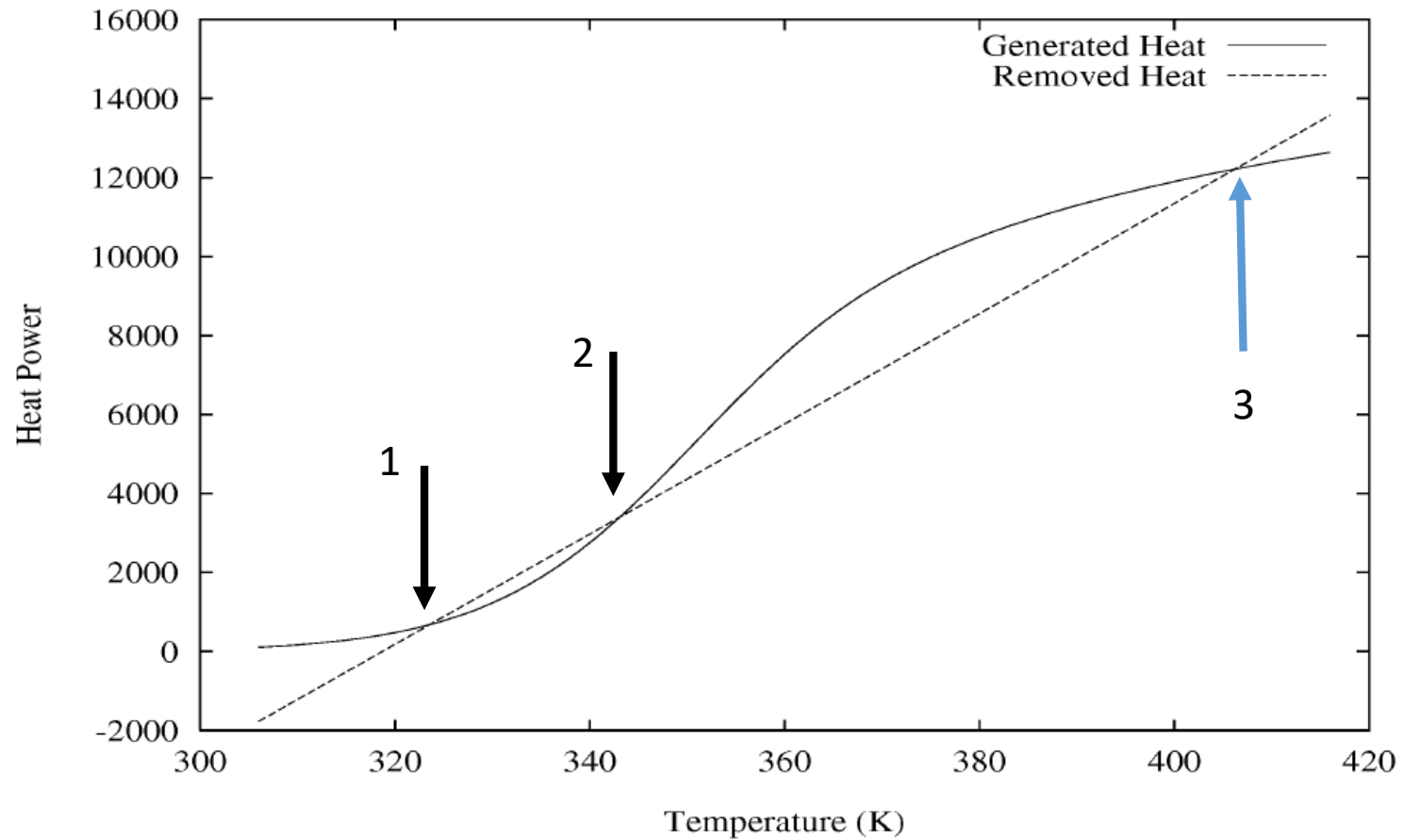
$$r = k_0 e^{-\frac{E}{RT}} C_A$$

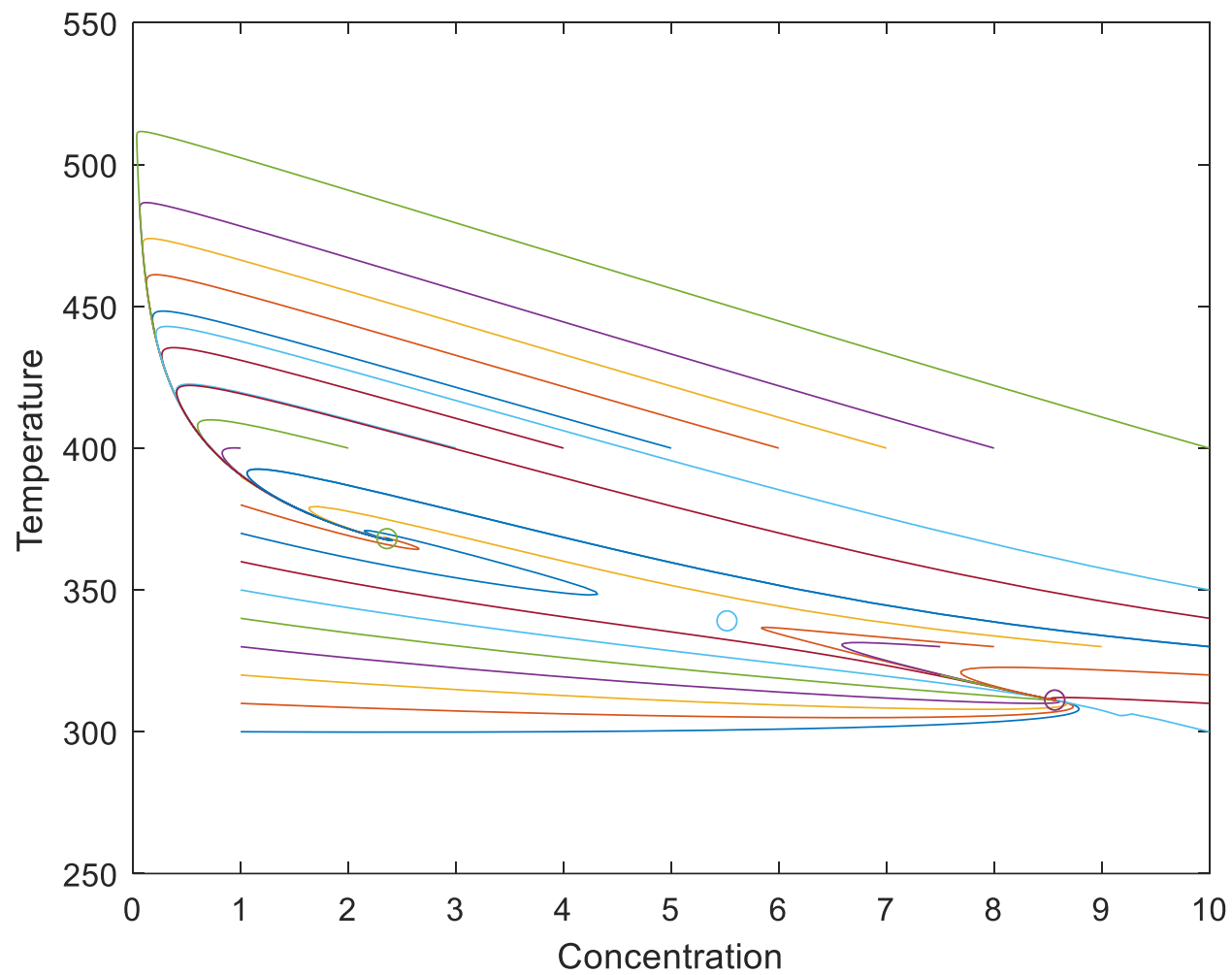
Case	$k_0, 1/\text{h}$	$(-\Delta H), \text{ kcal/kmol}$	UA/V
1	14825×3600	5215	250
2	9703×3600	5960	150
3	18194×3600	8195	750

Variation of Reactor Temp with Jacket Temperature



Steady State Heat Power vs Reactor Temperature





Developing Control Relevant Models

1. Develop dynamic model from 1st principles i.e,

$$\dot{x} = f(x, u) \quad \dots (1)$$

2. Define control objective $y = h(x, u) \dots\dots (2)$

3. Rearrange equation (1) and (2) to get control-affine nonlinear state space form

$$\begin{aligned}\dot{x} &= f(x) + g(x)u \\ y &= h(x, u)\end{aligned}$$

4. Linearize equation (1) and (2) around nominal values of x_s and u_s and form linear state space equation after subtracting steady state equation $f(x_s, u_s) = 0$; $h(x_s, u_s) = 0$

$$\begin{aligned}\dot{X} &= AX + BU \\ Y &= CX + DU\end{aligned}$$

Developing Control Relevant Models

5. Derive transfer function model by taking Laplace transform of linear state space model:

$$Y(s) = G_P(s)M(s) + G_L(s)L(s)$$

6. Put $s=j\omega$ and convert Laplace domain model to Frequency Domain transfer function model

$$Y(\omega) = G_P(\omega)M(\omega) + G_L(\omega)L(\omega)$$

7. Take inversion of Laplace domain transfer function model and get Convolution model

$$Y(t) = \int_0^t G_p(t - \tau) M(\tau) d\tau + \int_0^t G_L(t - \tau) L(\tau) d\tau$$

$$\begin{aligned}\dot{X} &= AX + BU \\ Y &= CX + DU\end{aligned}$$

Taking Laplace Transform,

$$\begin{aligned}sX(s) &= AX(s) + BU(s) \\ Y(s) &= CX(s) + DU(s)\end{aligned}$$

Or,

$$(sI - A)X(s) = BU(s); i.e, X(s) = (sI - A)^{-1}BU(s)$$

So,

$$Y(s) = [C(sI - A)^{-1}B + D]U(s)$$

Normally transfer function model is expressed in terms of process and disturbance transfer function. So, input variables U are partitioned to manipulated M and load/disturbance variable L i.e, $U = [M \ L]$

$$Y(s) = [C(sI - A)^{-1}B_M + D_M]M(s) + [C(sI - A)^{-1}B_L + D_L]L(s)$$

$$Y(s) = G_P(s)M(s) + G_L(s)L(s)$$

Dynamic model:

$$\dot{x} = f(x, u) \quad \dots (1)$$

$$y = h(x, u) \quad \dots (2)$$

Using Taylor series approximation around (x^s, u^s) and neglecting HOT

$$\dot{x}_1 = f_1(x_1^s, x_2^s, \dots, x_n^s, u_1^s, u_2^s, \dots, u_m^s) +$$

$$\frac{\partial f_1}{\partial x_1}(x_1 - x_1^s) + \frac{\partial f_1}{\partial x_2}(x_2 - x_2^s) + \dots + \frac{\partial f_1}{\partial x_n}(x_n - x_n^s)$$

$$+ \frac{\partial f_1}{\partial u_1}(u_1 - u_1^s) + \frac{\partial f_1}{\partial u_2}(u_2 - u_2^s) + \dots + \frac{\partial f_1}{\partial u_m}(u_m - u_m^s)$$

.....

.....

.....

$$\dot{x}_n = f_n(x_1^s, x_2^s, \dots, x_n^s, u_1^s, u_2^s, \dots, u_m^s) +$$

$$\frac{\partial f_n}{\partial x_1}(x_1 - x_1^s) + \frac{\partial f_n}{\partial x_2}(x_2 - x_2^s) + \dots + \frac{\partial f_n}{\partial x_n}(x_n - x_n^s)$$

$$+ \frac{\partial f_n}{\partial u_1}(u_1 - u_1^s) + \frac{\partial f_n}{\partial u_2}(u_2 - u_2^s) + \dots + \frac{\partial f_n}{\partial u_m}(u_m - u_m^s)$$

Subtracting the steady state equations, we can write:

$$\begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \\ \vdots \\ \dot{X}_n \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} + \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \dots & \frac{\partial f_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \dots & \frac{\partial f_n}{\partial u_m} \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ \vdots \\ U_m \end{bmatrix}$$

i.e,

$$\dot{X} = AX + BU$$

Where,

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \quad a_{ij} = \frac{\partial f_i}{\partial x_j} \quad \text{and} \quad B = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \dots & \frac{\partial f_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \dots & \frac{\partial f_n}{\partial u_m} \end{bmatrix} \quad b_{ij} = \frac{\partial f_i}{\partial u_j}$$

Similarly for equation (2),

$$y_1 = h_1(x_1^s, x_2^s, \dots \dots x_n^s, u_1^s, u_2^s, \dots \dots u_m^s) +$$

$$\frac{\partial h_1}{\partial x_1}(x_1 - x_1^s) + \frac{\partial h_1}{\partial x_2}(x_2 - x_2^s) + \dots \dots + \frac{\partial h_1}{\partial x_n}(x_n - x_n^s)$$

$$+ \frac{\partial h_1}{\partial u_1}(u_1 - u_1^s) + \frac{\partial h_1}{\partial u_2}(u - u_2^s) + \dots \dots + \frac{\partial h_1}{\partial u_m}(u_m - u_m^s)$$

.....

.....

.....

$$y_p = h_p(x_1^s, x_2^s, \dots \dots x_n^s, u_1^s, u_2^s, \dots \dots u_m^s) +$$

$$\frac{\partial h_p}{\partial x_1}(x_1 - x_1^s) + \frac{\partial h_p}{\partial x_2}(x_2 - x_2^s) + \dots \dots + \frac{\partial h_p}{\partial x_n}(x_n - x_n^s)$$

$$+ \frac{\partial h_p}{\partial u_1}(u_1 - u_1^s) + \frac{\partial h_p}{\partial u_2}(u - u_2^s) + \dots \dots + \frac{\partial h_p}{\partial u_m}(u_m - u_m^s)$$

Subtracting the steady state equations, we can write:

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_p \end{bmatrix} = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \dots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial x_1} & \dots & \frac{\partial h_p}{\partial x_n} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} + \begin{bmatrix} \frac{\partial h_1}{\partial u_1} & \dots & \frac{\partial h_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial u_1} & \dots & \frac{\partial h_p}{\partial u_m} \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ \vdots \\ U_m \end{bmatrix}$$

i.e,

$$Y = CX + DU$$

Where,

$$C = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \dots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial x_1} & \dots & \frac{\partial h_p}{\partial x_n} \end{bmatrix} \quad c_{ij} = \frac{\partial h_i}{\partial x_j} \quad \text{and} \quad D = \begin{bmatrix} \frac{\partial h_1}{\partial u_1} & \dots & \frac{\partial h_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial u_1} & \dots & \frac{\partial h_p}{\partial u_m} \end{bmatrix} \quad d_{ij} = \frac{\partial h_i}{\partial u_j}$$

Example: Van De Vusse Reactor

Dynamic Model:

$$\frac{dC_A}{dt} = \frac{F}{V} (C_{Af} - C_A) - k_1 C_A - k_3 C_A^2 = f_1(C_A, C_B, F/V)$$

$$\frac{dC_B}{dt} = -\frac{F}{V} C_B + k_1 C_A - k_2 C_B = f_2(C_A, C_B, F/V)$$

Non-linear state space model:

States are : $x_1 = C_A$, $x_2 = C_B$ input : $u = F/V$ and output $y = C_B$

State equation:

$$\frac{dx_1}{dt} = (-k_1 x_1 - k_3 x_1^2) + (x_{1f} - x_1)u$$

$$\frac{dx_2}{dt} = k_1 x_1 - k_2 x_2 + (-x_2)u$$

$$\text{i.e, } \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{Bmatrix} (-k_1 x_1 - k_3 x_1^2) \\ k_1 x_1 - k_2 x_2 \end{Bmatrix} + \begin{Bmatrix} (x_{1f} - x_1) \\ (-x_2) \end{Bmatrix} u = f(x) + g(x)u$$

$$\text{Output map: } y = h(x) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Example: Van De Vusse Reactor

Linear State Space Model:

$$f_1(C_A, C_B, F/V) = \frac{F}{V} (C_{Af} - C_A) - k_1 C_A - k_3 C_A^2$$

$$f_2(C_A, C_B, F/V) = -\frac{F}{V} C_B + k_1 C_A - k_2 C_B$$

$$\text{Let, } X_1 = C_A - C_A^S; X_2 = C_B - C_B^S; U = \frac{F}{V} - \frac{F^S}{V}$$

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial C_A} & \frac{\partial f_1}{\partial C_B} \\ \frac{\partial f_2}{\partial C_A} & \frac{\partial f_2}{\partial C_B} \end{bmatrix} = \begin{bmatrix} -F^S/V - k_1 - 2k_3 C_A^S & 0 \\ k_1 & -F^S/V - k_2 \end{bmatrix}$$

$$B = \begin{bmatrix} \frac{\partial f_1}{\partial (\frac{F}{V})} \\ \frac{\partial f_2}{\partial (\frac{F}{V})} \end{bmatrix} = \begin{bmatrix} C_{Af} - C_A^S \\ -C_B^S \end{bmatrix} \quad C = [0 \quad 1] \quad D = 0$$

$$\dot{X} = AX + BU$$

$$Y = CX + DU$$

Example: Van De Vusse Reactor

Transfer Domain Model -- Laplace Domain:

$$Y(s) = [C(sI - A)^{-1}B + D]U(s) = [C(sI - A)^{-1}B]U(s)$$

$$sI - A = \begin{bmatrix} s + F^s/V + k_1 + 2k_3C_A^s & 0 \\ -k_1 & s + F^s/V + k_2 \end{bmatrix} = \begin{bmatrix} s - a_{11} & -a_{12} \\ -a_{21} & s - a_{22} \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{\det} \begin{bmatrix} s - a_{22} & a_{12} \\ a_{21} & s - a_{11} \end{bmatrix}$$

where, $\det = (s - a_{11})(s - a_{22}) - a_{12}a_{21}$

$$C(sI - A)^{-1} = [0 \quad 1] \frac{1}{\det} \begin{bmatrix} s - a_{22} & a_{12} \\ a_{21} & s - a_{11} \end{bmatrix} = \frac{1}{\det} [a_{21} \quad s - a_{11}]$$

$$C(sI - A)^{-1}B = \frac{1}{\det} [a_{21} \quad s - a_{11}] \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \frac{a_{21}b_1 + (s - a_{11})b_2}{(s - a_{11})(s - a_{22}) - a_{12}a_{21}}$$

Example: Van De Vusse Reactor

Transfer Domain Model -- Laplace Domain:

$$\begin{aligned} C(sI - A)^{-1}B &= \frac{a_{21}b_1 + (s - a_{11})b_2}{(s - a_{11})(s - a_{22}) - a_{12}a_{21}} \\ &= \frac{k_1(C_{Af} - C_A^s) - C_B^s s - C_B^s (F^s/V + k_1 + 2k_3 C_A^s)}{(s + F^s/V + k_1 + 2k_3 C_A^s)(s + F^s/V + k_2)} \\ &= \frac{-C_B^s s + [k_1(C_{Af} - C_A^s) - C_B^s (F^s/V + k_1 + 2k_3 C_A^s)]}{(s + F^s/V + k_1 + 2k_3 C_A^s)(s + F^s/V + k_2)} \end{aligned}$$

Process Dynamics & Control

Control relevant models

Control relevant models are often quite simple compared to the true process and usually combine physical reasoning with experimental data.

State Space Models – Non-linear

➤ Continuous Time Systems

$$\begin{aligned}\frac{dx}{dt} &= f(x(t), u(t), t) \\ y(t) &= h(x(t), u(t), t)\end{aligned}$$

Control affine form

$$\begin{aligned}\frac{dx}{dt} &= f(x) + g(x)u \\ y &= h(x)\end{aligned}$$

➤ Discrete Time System

$$\begin{aligned}x_{k+1} &= f_d(x_k, u_k, k) \\ y_k &= h_d(x_k, u_k, k)\end{aligned}$$

$$\begin{aligned}x_{k+1} &= f_d(x_k) + g_d(x_k)u_k \\ y_k &= h_d(x_k)\end{aligned}$$

x, u, y, f, g, h can be vectors depending of number of inputs and outputs

Control relevant models

State Space Models – linear

➤ Continuous Time Systems

$$\frac{dX}{dt} = \mathbf{A}X(t) + \mathbf{B}U(t)$$

$$Y = \mathbf{C}X(t) + \mathbf{D}U(t)$$

Discrete Time Systems

$$X_{k+1} = \mathbf{A}_d X_k + \mathbf{B}_d U_k$$

$$Y_k = \mathbf{C}_d X_k + \mathbf{D}_d U_k$$

X,Y,U are in deviation form and can be vector depending on the input / output of the system

Control relevant models

Transform Domain Models

➤ Laplace S-domain

$$Y(s) = G(s)U(s) = G_p(s)M(s) + G_L(s)L(s)$$

➤ Frequency domain

$$Y(j\omega) = G_p(j\omega)M(j\omega) + G_L(j\omega)L(j\omega)$$

➤ Discrete time Z-domain

$$Y(z) = G(z)U(z) = G_p(z)M(z) + G_L(z)L(z)$$

Control relevant models

Convolution Models

➤ Continuous Time

$$Y(t) = \int_0^t G(\tau)U(t - \tau)d\tau = \int_0^t U(\tau)G(t - \tau)d\tau$$

$$Y(t) = \int_0^t G_p(\tau)M(t - \tau)d\tau + \int_0^t G_L(\tau)L(t - \tau)d\tau$$

➤ Discrete Time

$$Y(n) = \sum_0^n G_p(k) M(n - k) + \sum_0^n G_L(k) L(n - k)$$

Developing Control Relevant Models

1. Develop dynamic model from 1st principles i.e,

$$\dot{x} = f(x, u) \quad \dots (1)$$

2. Define control objective $y = h(x, u) \dots\dots (2)$

3. Rearrange equation (1) and (2) to get control-affine nonlinear state space form

$$\dot{x} = f(x) + g(x)u$$

$$y = h(x)$$

4. Linearize equation (1) and (2) around nominal values of x_s and u_s and form linear state space equation after subtracting steady state equation $f(x_s, u_s) = 0$; $h(x_s, u_s) = 0$

$$\dot{X} = AX + BU$$

$$Y = CX + DU$$

Developing Control Relevant Models

5. Derive transfer function model by taking Laplace transform of linear state space model:

$$Y(s) = G_P(s)M(s) + G_L(s)L(s)$$

6. Put $s=j\omega$ and convert Laplace domain model to Frequency Domain transfer function model

$$Y(\omega) = G_P(\omega)M(\omega) + G_L(\omega)L(\omega)$$

7. Take inversion of Laplace domain transfer function model and get Convolution model

$$Y(t) = \int_0^t G_p(t - \tau) M(\tau) d\tau + \int_0^t G_L(t - \tau) L(\tau) d\tau$$

Transfer function Model from Dynamic Equation

Dynamic model:

$$\dot{x} = f(x, u) \quad \dots (1)$$

$$y = h(x, u) \quad \dots (2)$$

Using Taylor series approximation around (x^s, u^s) and neglecting HOT

$$\dot{x}_1 = f_1(x_1^s, x_2^s, \dots, x_n^s, u_1^s, u_2^s, \dots, u_m^s) +$$

$$\frac{\partial f_1}{\partial x_1}(x_1 - x_1^s) + \frac{\partial f_1}{\partial x_2}(x_2 - x_2^s) + \dots + \frac{\partial f_1}{\partial x_n}(x_n - x_n^s)$$

$$+ \frac{\partial f_1}{\partial u_1}(u_1 - u_1^s) + \frac{\partial f_1}{\partial u_2}(u_2 - u_2^s) + \dots + \frac{\partial f_1}{\partial u_m}(u_m - u_m^s)$$

.....

.....

.....

$$\dot{x}_n = f_n(x_1^s, x_2^s, \dots, x_n^s, u_1^s, u_2^s, \dots, u_m^s) +$$

$$\frac{\partial f_n}{\partial x_1}(x_1 - x_1^s) + \frac{\partial f_n}{\partial x_2}(x_2 - x_2^s) + \dots + \frac{\partial f_n}{\partial x_n}(x_n - x_n^s)$$

$$+ \frac{\partial f_n}{\partial u_1}(u_1 - u_1^s) + \frac{\partial f_n}{\partial u_2}(u_2 - u_2^s) + \dots + \frac{\partial f_n}{\partial u_m}(u_m - u_m^s)$$

Subtracting the steady state equations, we can write:

$$\begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \\ \vdots \\ \dot{X}_n \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} + \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \dots & \frac{\partial f_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \dots & \frac{\partial f_n}{\partial u_m} \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ \vdots \\ U_m \end{bmatrix}$$

i.e,

$$\dot{X} = AX + BU$$

Where,

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \quad a_{ij} = \frac{\partial f_i}{\partial x_j} \quad \text{and} \quad B = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \dots & \frac{\partial f_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \dots & \frac{\partial f_n}{\partial u_m} \end{bmatrix} \quad b_{ij} = \frac{\partial f_i}{\partial u_j}$$

Similarly for equation (2),

$$y_1 = h_1(x_1^s, x_2^s, \dots \dots x_n^s, u_1^s, u_2^s, \dots \dots u_m^s) +$$

$$\frac{\partial h_1}{\partial x_1}(x_1 - x_1^s) + \frac{\partial h_1}{\partial x_2}(x_2 - x_2^s) + \dots \dots + \frac{\partial h_1}{\partial x_n}(x_n - x_n^s)$$

$$+ \frac{\partial h_1}{\partial u_1}(u_1 - u_1^s) + \frac{\partial h_1}{\partial u_2}(u - u_2^s) + \dots \dots + \frac{\partial h_1}{\partial u_m}(u_m - u_m^s)$$

.....

.....

.....

$$y_p = h_p(x_1^s, x_2^s, \dots \dots x_n^s, u_1^s, u_2^s, \dots \dots u_m^s) +$$

$$\frac{\partial h_p}{\partial x_1}(x_1 - x_1^s) + \frac{\partial h_p}{\partial x_2}(x_2 - x_2^s) + \dots \dots + \frac{\partial h_p}{\partial x_n}(x_n - x_n^s)$$

$$+ \frac{\partial h_p}{\partial u_1}(u_1 - u_1^s) + \frac{\partial h_p}{\partial u_2}(u - u_2^s) + \dots \dots + \frac{\partial h_p}{\partial u_m}(u_m - u_m^s)$$

Subtracting the steady state equations, we can write:

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_p \end{bmatrix} = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \dots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial x_1} & \dots & \frac{\partial h_p}{\partial x_n} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} + \begin{bmatrix} \frac{\partial h_1}{\partial u_1} & \dots & \frac{\partial h_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial u_1} & \dots & \frac{\partial h_p}{\partial u_m} \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ \vdots \\ U_m \end{bmatrix}$$

i.e,

$$Y = CX + DU$$

Where,

$$C = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \dots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial x_1} & \dots & \frac{\partial h_p}{\partial x_n} \end{bmatrix} \quad c_{ij} = \frac{\partial h_i}{\partial x_j} \quad \text{and} \quad D = \begin{bmatrix} \frac{\partial h_1}{\partial u_1} & \dots & \frac{\partial h_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial u_1} & \dots & \frac{\partial h_p}{\partial u_m} \end{bmatrix} \quad d_{ij} = \frac{\partial h_i}{\partial u_j}$$

$$\begin{aligned}\dot{X} &= AX + BU \\ Y &= CX + DU\end{aligned}$$

Taking Laplace Transform,

$$\begin{aligned}sX(s) &= AX(s) + BU(s) \\ Y(s) &= CX(s) + DU(s)\end{aligned}$$

Or,

$$(sI - A)X(s) = BU(s); i.e, X(s) = (sI - A)^{-1}BU(s)$$

So,

$$Y(s) = [C(sI - A)^{-1}B + D]U(s)$$

Normally transfer function model is expressed in terms of process and disturbance transfer function. So, input variables U are partitioned to manipulated M and load/disturbance variable L i.e, $U = [M \ L]$

$$Y(s) = [C(sI - A)^{-1}B_M + D_M]M(s) + [C(sI - A)^{-1}B_L + D_L]L(s)$$

$$Y(s) = G_P(s)M(s) + G_L(s)L(s)$$

Example: Van De Vusse Reactor

Dynamic Model:

$$\frac{dC_A}{dt} = \frac{F}{V} (C_{Af} - C_A) - k_1 C_A - k_3 C_A^2 = f_1(C_A, C_B, F/V)$$

$$\frac{dC_B}{dt} = -\frac{F}{V} C_B + k_1 C_A - k_2 C_B = f_2(C_A, C_B, F/V)$$

Non-linear state space model:

States are : $x_1 = C_A$, $x_2 = C_B$ input : $u = F/V$ and output $y = C_B$

State equation:

$$\frac{dx_1}{dt} = (-k_1 x_1 - k_3 x_1^2) + (x_{1f} - x_1)u$$

$$\frac{dx_2}{dt} = k_1 x_1 - k_2 x_2 + (-x_2)u$$

$$\text{i.e, } \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{Bmatrix} (-k_1 x_1 - k_3 x_1^2) \\ k_1 x_1 - k_2 x_2 \end{Bmatrix} + \begin{Bmatrix} (x_{1f} - x_1) \\ (-x_2) \end{Bmatrix} u = f(x) + g(x)u$$

Output map:

$$y = h(x) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Example: Van De Vusse Reactor

Linear State Space Model:

$$\dot{C}_A = f_1(C_A, C_B, F/V) = \frac{F}{V}(C_{Af} - C_A) - k_1 C_A - k_3 C_A^2$$

$$\dot{C}_B = f_2(C_A, C_B, F/V) = -\frac{F}{V} C_B + k_1 C_A - k_2 C_B$$

Let, $X_1 = C_A - C_A^S$; $X_2 = C_B - C_B^S$; $U = \frac{F}{V} - \frac{F^S}{V}$

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial C_A} & \frac{\partial f_1}{\partial C_B} \\ \frac{\partial f_2}{\partial C_A} & \frac{\partial f_2}{\partial C_B} \end{bmatrix} = \begin{bmatrix} -F^S/V - k_1 - 2k_3 C_A^S & 0 \\ k_1 & -F^S/V - k_2 \end{bmatrix}$$

$$B = \begin{bmatrix} \frac{\partial f_1}{\partial (\frac{F}{V})} \\ \frac{\partial f_2}{\partial (\frac{F}{V})} \end{bmatrix} = \begin{bmatrix} C_{Af} - C_A^S \\ -C_B^S \end{bmatrix}$$

$$C = [0 \quad 1] \quad D = 0$$

$$\dot{X} = \begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \end{bmatrix} = AX + BU$$

$$Y = CX + DU$$

Example: Van De Vusse Reactor

Transfer Domain Model -- Laplace Domain:

$$Y(s) = [C(sI - A)^{-1}B + D]U(s) = [C(sI - A)^{-1}B]U(s)$$

$$sI - A = \begin{bmatrix} s + F^s/V + k_1 + 2k_3C_A^s & 0 \\ -k_1 & s + F^s/V + k_2 \end{bmatrix} = \begin{bmatrix} s - a_{11} & -a_{12} \\ -a_{21} & s - a_{22} \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{\det} \begin{bmatrix} s - a_{22} & a_{12} \\ a_{21} & s - a_{11} \end{bmatrix}$$

where, $\det = (s - a_{11})(s - a_{22}) - a_{12}a_{21}$

$$C(sI - A)^{-1} = [0 \quad 1] \frac{1}{\det} \begin{bmatrix} s - a_{22} & a_{12} \\ a_{21} & s - a_{11} \end{bmatrix} = \frac{1}{\det} [a_{21} \quad s - a_{11}]$$

$$C(sI - A)^{-1}B = \frac{1}{\det} [a_{21} \quad s - a_{11}] \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \frac{a_{21}b_1 + (s - a_{11})b_2}{(s - a_{11})(s - a_{22}) - a_{12}a_{21}}$$

Example: Van De Vusse Reactor

Transfer Domain Model -- Laplace Domain:

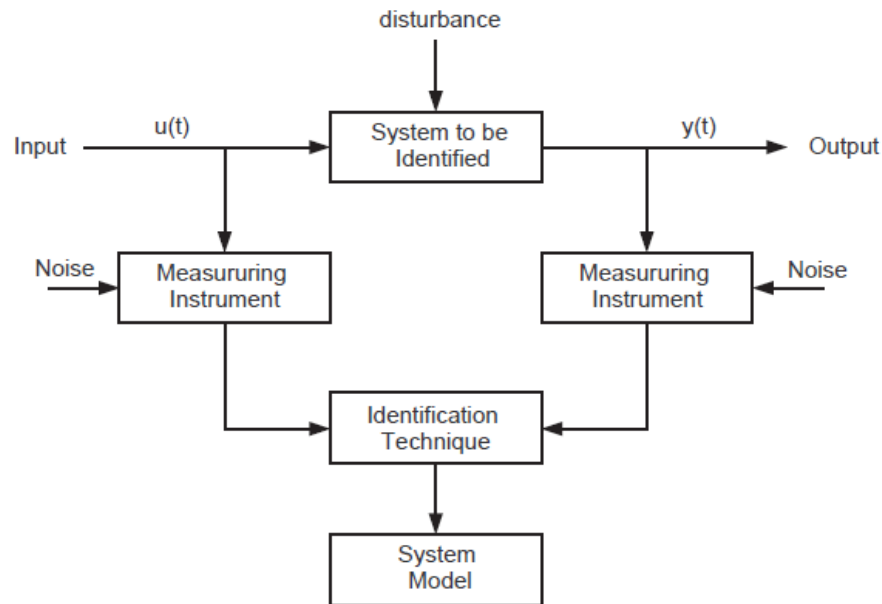
$$\begin{aligned} C(sI - A)^{-1}B &= \frac{a_{21}b_1 + (s - a_{11})b_2}{(s - a_{11})(s - a_{22}) - a_{12}a_{21}} \\ &= \frac{k_1(C_{Af} - C_A^s) - C_B^s s - C_B^s (F^s/V + k_1 + 2k_3 C_A^s)}{(s + F^s/V + k_1 + 2k_3 C_A^s)(s + F^s/V + k_2)} \\ &= \frac{-C_B^s s + [k_1(C_{Af} - C_A^s) - C_B^s (F^s/V + k_1 + 2k_3 C_A^s)]}{(s + F^s/V + k_1 + 2k_3 C_A^s)(s + F^s/V + k_2)} \end{aligned}$$

Process Dynamics & Control

Process Identification

What is Identification?

Identification is the process of constructing an empirical model of a (dynamical) system from observations of its inputs and outputs. In a dynamical system, the output at any instant depends on its history not just present input, i.e. it has some form of memory (storage).



Classification of identification

Based on the degree of a priori knowledge of the system.

Black Box:

- This means we know nothing about the basic characteristics of the system.
- Extremely difficult to solve.
- Usually some kind of assumptions have to be made before any meaningful solution can be attempted.

Grey Box:

- In this case, some basic characteristics of the system are known (ie. linearity, bandwidth, structure).
- However, order of the dynamic equation or values of the associated coefficients may be unknown.

Process Identification Procedure

The experiment design objective is to make choices such that the collected data is maximally informative, i.e.

- types of signals,
- amplitudes,
- what to measure
- input/outputs

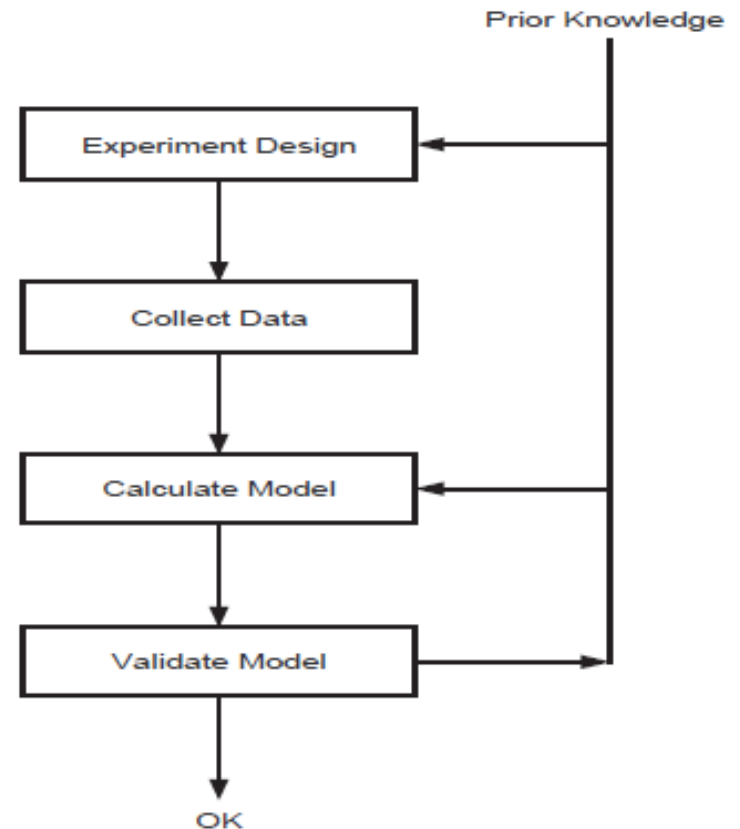
In the model calculation block, decisions need to be made on:

- what type of model,
- model structure,
- parameter estimation technique

Process Identification Procedure

To validate the model, the following questions need to be asked:

- How does the model output relate to the data observed?
- Is it good enough for specific purpose?



Types of Test Signals

The input signal used in an identification experiment can have a significant influence on the resulting parameter estimates.

Typical types of input signal often used in practice include:

- Impulse function (approximate)
- Step function
- Pseudo-random binary sequence
- Sinusoids

The signal we consider in this course is step function.

Types of Test Signals

Step function

- From the response of a process to a step input a number of practical parameters can be obtained, i.e. dead time, time constant, etc.
- It is easy to apply.
- It is very sensitive to noise. (Usually requires a rather large amplitude to obtain reasonable results)
- Generally, it can only give a basic model.
- Steady state gain is easily found, in the absence of drift, from the initial and final values of the step response.
- User choices to be considered:
 - Amplitude
 - Duration

Design of Identification Experiments

There are a number of factors to consider before performing an identification experiment.

- What form of test signal should be used, i.e. step, doublet, prbs?
 - This depends on the quality of the model you require. In most cases, for PID control, step tests are adequate.
- What size should the amplitude be? There are quite a number of factors to consider here.
 - There may be constraints on how much variation can be tolerated in the input and/or output. (economic, safety, actuator limits, etc).
 - One reason for a large amplitude is that the effects of noise become less (signal to noise ratio is larger).
 - For systems with known nonlinearities it is best to keep the amplitude small as generally you are interested in a model around a particular operating point.

Design of Identification Experiments

- What sampling frequency should be used?
 - Typically, the sampling frequency should be chosen as 10 - 20× that of the test signal frequency. (5× should be considered as the absolute minimum).
 - For a step test, you want to capture at least 5 samples per the time constant of the process.
- What should the frequencies of multi-step test signal be?
 - The frequencies of these test signals should be in the frequency region of interest and should be chosen as a multiple of the sampling frequency and of each other. This will minimise errors.
 - When collecting data from a multi-step test one should wait until the transients have decayed significantly.

Transfer Function from Process Data

- Normally, chemical process transfer functions are approximated by First Order with Dead Time

(FODT) model,
$$g(s) = \frac{K e^{-\theta s}}{\tau s + 1}$$

- Four classical methods (using step response data)
 - Ziegler-Nichols method
 - Smith's method
 - Sundaresan and Krishnaswamy method
 - Nishikawa's method

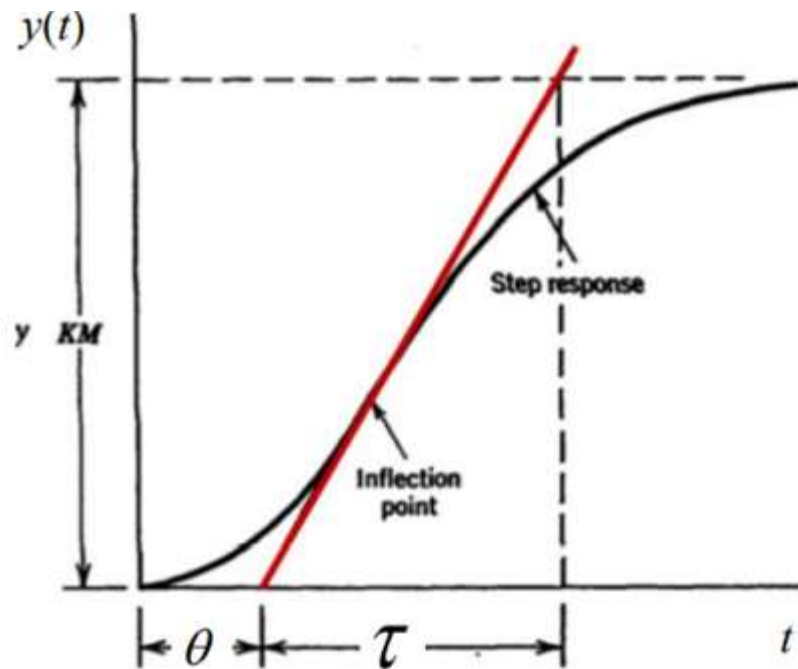
Transfer function from Process Data

Ziegler-Nichols method (1942) consists of applying a tangent line to the curve at an inflection point, to determine the system's gain, time, and delay constants, as shown in Figure.

The inflection point is defined where the curve changes direction and the derivative is equal to zero.

Drawback:

Difficult to find inflection Point and to draw tangent.



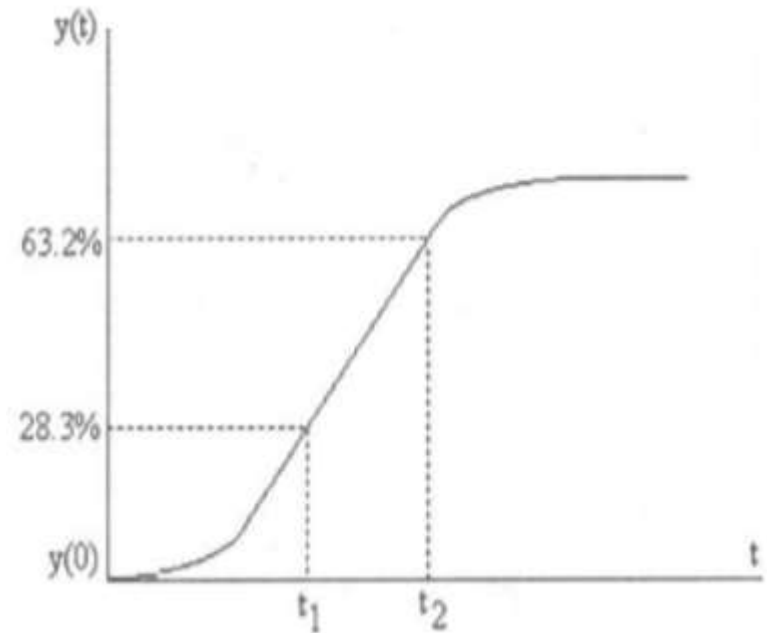
Transfer function from Process Data

Smith's Method (1972) proposed that the values of θ and τ be selected in such a way that the model and the real responses coincide in two points that present a high rate of variation.

The two time points are at 28.3% 63.2 % of change in output respectively.

So from figure, after finding t_1 and t_2 , we can calculate

$$K = \frac{\Delta output}{\Delta input} \quad \tau = \frac{3}{2}(t_2 - t_1) \quad \text{and} \quad \theta = t_2 - \tau$$



Transfer function from Process Data

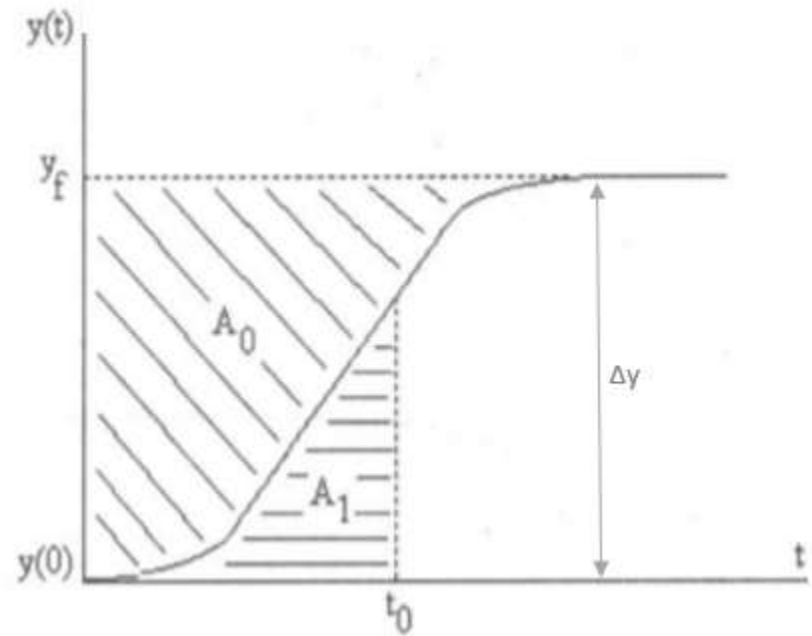
Sundaresan and Krishnaswamy Method:

This is similar to smith's method but the two time points t_1 and t_2 are based on 35.3% and 85.3% of change in output response respectively. So, model parameters are calculated as,

$$K = \frac{\Delta output}{\Delta input} ; \tau = \frac{2}{3}(t_2 - t_1); \theta = 1.3t_1 - 0.29t_2$$

Transfer function from Process Data

Nishikawa (2007) method determine the values of the constants using the calculation of the curve areas as shown in Figure.

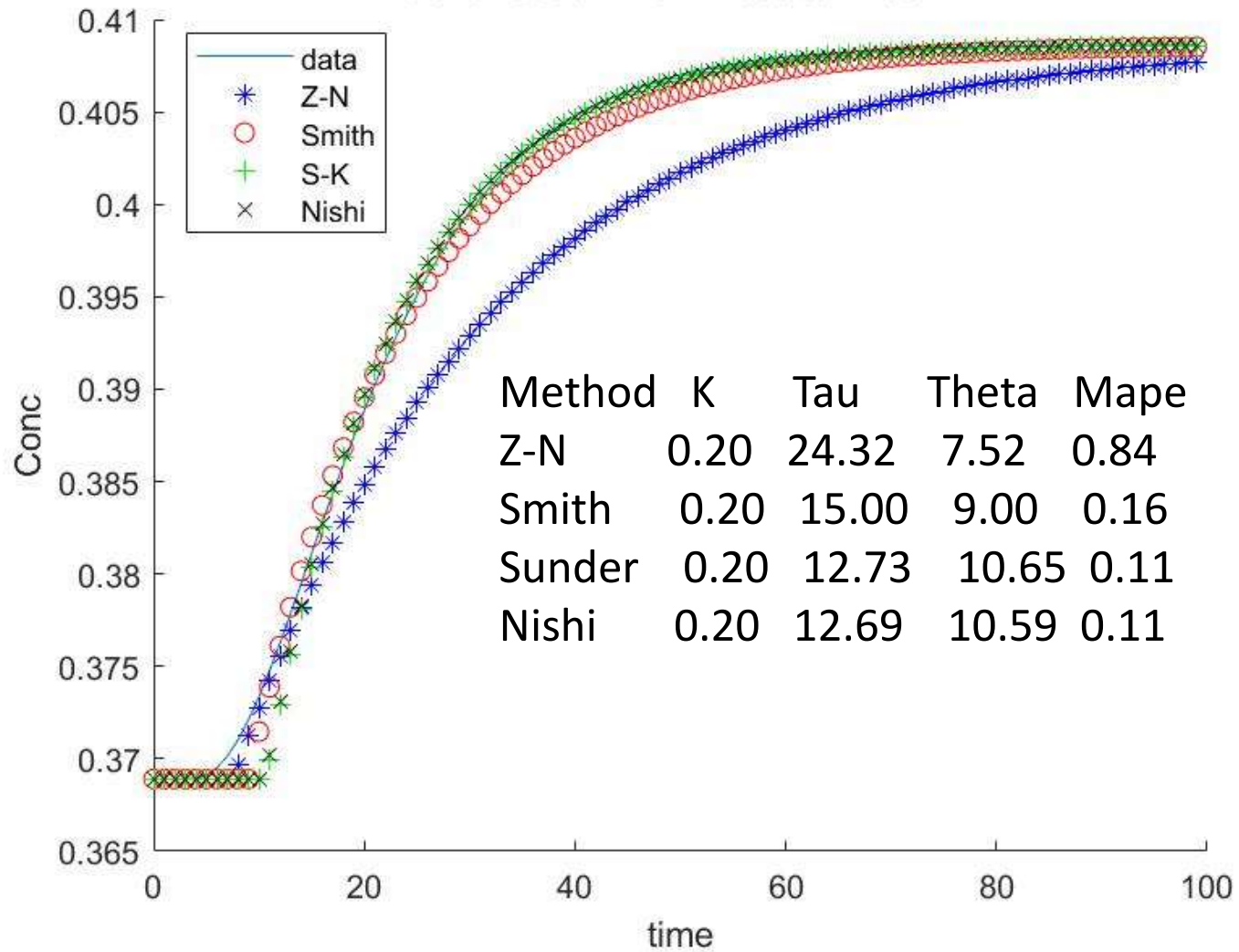


$$A_0 = \int_0^{\infty} (\Delta y(\infty) - \Delta y(t)) dt$$

$$A_1 = \int_0^{t_0} \Delta y(t) dt \text{ where, } t_0 = \frac{A_0}{\Delta y(\infty)}$$

$$\tau = \frac{A_1}{0.368 \Delta y(\infty)} ; \theta = t_0 - \tau ; K = \frac{\Delta y(\infty)}{\Delta c(\infty)}$$

Identification from Reactor Data



Transfer function from Process Data

Method of Moments method

- can be used to determine model parameters from the output response generated using arbitrary input function.
- is based on the definition of the Laplace transform of the impulse response $g(t)$ of a system, which is its transfer function $G(s) = \int_0^{\infty} e^{-st} g(t) dt$

As the n-th-order moment of a function $f(x)$ is defined by

$$M_n(f) = \int_0^{\infty} x^n f(x) dx$$

Transfer function from Process Data

It can be noticed that the first two derivatives of $G(s)$ w.r.t s
 $G'(s) = -\int_0^\infty t e^{-st} g(t) dt$ & $G''(s) = \int_0^\infty t^2 e^{-st} g(t) dt$
are related to the moments of the impulse response function
by $G(0) = \int_0^\infty g(t) dt$; $G'(0) = -\int_0^\infty t g(t) dt$; and
 $G''(0) = \int_0^\infty t^2 g(t) dt$;

- Thus $G(0)$, $-G'(0)$, $G''(0)$ are respectively the zero-, first- and second-order moments of the impulse response $g(t)$.
- Note that the above three integrals can be calculated by using the measured output response.

Transfer function from Process Data

- Note that in this method, it is possible to use any type of input. We have, in general

$$Y(s) = G(s) U(s) ; Y'(s) = G'(s) U(s) + G(s) U'(s)$$

$$Y''(s) = G''(s) U(s) + G(s) U''(s) + 2 G'(s) U'(s)$$

- from which the following equations are deduced

$$Y(0) = G(0) U(0) ; Y'(0) = G'(0) U(0) + G(0) U'(0)$$

$$Y''(0) = G''(0) U(0) + G(0) U''(0) + 2 G'(0) U'(0)$$

These quantities can be calculated by the following equations:

$$U(0) = \int_0^{\infty} u(t) dt ; U'(0) = - \int_0^{\infty} t u(t) dt ;$$

$$U''(0) = \int_0^{\infty} t^2 u(t) dt ; Y(0) = \int_0^{\infty} y(t) dt ;$$

$$Y'(0) = - \int_0^{\infty} t y(t) dt ; \text{ and } Y''(0) = \int_0^{\infty} t^2 y(t) dt ;$$

Transfer function from Process Data

FODT model: $G(s) = \frac{K e^{-\theta s}}{\tau s + 1}$

So, $G(0) = K = \frac{Y(0)}{U(0)}$

$$G'(0) = -K(\tau + \theta) = \frac{Y'(0) - G(0)U'(0)}{U(0)}$$

$$G''(0) = K(2\tau^2 + 2\tau\theta + \theta^2) = \frac{Y''(0) - G(0)U''(0) - 2G'(0)U'(0)}{U(0)}$$

Solving above equations K, τ, θ can be obtained.

SODT model from step response

- $G(s) = \frac{K e^{-\theta s}}{\tau^2 s^2 + 2\xi\tau s + 1}$ with $\xi > 0.707$

Method (Rangaiah & Krishnaswamy)

1. Calculate the process steady-state gain from the magnitude of the step input and that of the corresponding response;
2. Find t_1 , t_2 and t_3 from the response data corresponding to 14%, 55%, and 91% of the actual response;

SODT model from step response

3. Calculate α , β and ξ using the following equations

$$\alpha = \frac{t_3 - t_2}{t_2 - t_1}$$
$$\beta = \ln \left(\frac{\alpha}{2.485 - \alpha} \right)$$

$$\xi^2 = 0.50906 + 0.51743\beta - 0.076284\beta^2 + 0.041363\beta^3 \\ - 0.0049224\beta^4 + 0.00021234\beta^5$$

for $1.2323 < \alpha < 2.485$

SODT model from step response

4. Calculate τ and θ using the following equations

$$\frac{t_2 - t_1}{\tau} = 0.85818 - 0.62907\xi + 1.2897 \xi^2 - 0.36859 \xi^3 + 0.038891 \xi^4$$

$$\frac{t_2 - \theta}{\tau} = 1.392 - 0.52536 \xi + 1.2991 \xi^2 - 0.36014 \xi^3 + 0.037605 \xi^4$$

Non-linear Regression

This method can be used for any form of linear or non-linear models.

For example, process having transfer function of one zero 2 pole system

$$\frac{y(s)}{u(s)} = G(s) = \frac{K(\tau_3 s + 1)}{(\tau_1 s + 1)(\tau_2 s + 1)}$$

For Step input of magnitude M in u gives,

$$y(t) = KM \left(1 + \frac{\tau_3 - \tau_1}{\tau_1 - \tau_2} e^{-\frac{t}{\tau_1}} + \frac{\tau_3 - \tau_2}{\tau_2 - \tau_1} e^{-\frac{t}{\tau_2}} \right)$$

The above equation can be regressed to get the model parameters $K, \tau_1, \tau_2, \tau_3$

Matlab Implementation

Step response data:

t	0	1	2	3	4	5	6	7	8	9	10
u	1	1	1	1	1	1	1	1	1	1	1
y	0	0.058	0.217	0.36	0.488	0.6	0.692	0.772	0.833	0.888	0.925

Use command

```
mdl = procest(data, Type)
```

Where, data should stored in iddata format.

Type = 'P2DUZ' [2 poles, delay, U underdamped, Z zero]

```
>> u=[0;0;0;u]; y=[0;0;0;y]; % for step changes
```

```
>> z = iddata(y, u, 1); % y and u should be column vectors
```

```
>> mdl= procest(z,'P2DUZ');
```

'procest' can work for step as well as arbitrary input changes.

Process Dynamics & Control

State Space realization from Transfer Function

- Process of converting transfer function to state space form is not unique.
- Various realizations possible
- All realizations are equivalent
- One realization may have some advantages over others for a particular task

Possible realizations :

- First Companion form (Controllable Canonical Form)
- Jordan Canonical form
- Alternate first companion form (Toeplitz form)
- Second Companion form (Observable canonical form)

Controllable Canonical Form

Consider Laplace domain transfer function

$$g(s) = \frac{y(s)}{u(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$
$$= \frac{y(s)}{z(s)} \frac{z(s)}{u(s)} = (b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n) \left(\frac{1}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} \right)$$

Considering, $\frac{z(s)}{u(s)} = \frac{1}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$

i.e., $\frac{d^n z}{dt^n} + a_1 \frac{d^{n-1} z}{dt^{n-1}} + \dots + a_{n-1} \frac{dz}{dt} + a_n z = u$

Controllable Canonical Form

Let us choose the states x_1 to x_n as,

$$x_1 = z; x_2 = \frac{dz}{dt}; x_3 = \frac{d^2z}{dt^2} \dots \dots x_n = \frac{d^{n-1}z}{dt^{n-1}}$$

Therefore, the state equations are:

$$\dot{x}_1 = x_2;$$

$$\dot{x}_2 = x_3;$$

.

.

$$\dot{x}_{n-1} = x_n;$$

$$\dot{x}_n = -a_n x_1 - a_{n-1} x_2 - \dots \dots - a_2 x_{n-1} - a_1 x_n + u;$$

Controllable Canonical Form

Now for the output map,

$$\frac{y(s)}{z(s)} = b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n$$

$$\begin{aligned} \text{i.e, } y &= b_0 \frac{d^n z}{dt^n} + b_1 \frac{d^{n-1} z}{dt^{n-1}} + \dots + b_{n-1} \frac{dz}{dt} + b_n z \\ &= b_0 \dot{x}_n + b_1 x_n + b_2 x_{n-1} + \dots + b_{n-1} x_2 + b_n x_1 \\ &= b_0 (-a_n x_1 - a_{n-1} x_2 - \dots - a_2 x_{n-1} - a_1 x_n) \\ &\quad + b_1 x_n + b_2 x_{n-1} + \dots + b_{n-1} x_2 + b_n x_1 + b_0 u \\ &= (b_n - a_n b_0) x_1 + (b_{n-1} - a_{n-1} b_0) x_2 + \dots \\ &\quad + (b_2 - a_2 b_0) x_{n-1} + (b_1 - a_1 b_0) x_n + b_0 u \end{aligned}$$

Controllable Canonical Form

So, in standard vector-matrix form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdot & \cdot & 0 & 0 \\ 0 & 0 & 1 & 0 & \cdot & \cdot & 0 & 0 \\ 0 & 0 & 0 & 1 & \cdot & \cdot & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot & \cdot & 0 & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdot & \cdot & \cdot & -a_2 & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \cdot \\ \cdot \\ \cdot \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \cdot \\ \cdot \\ \cdot \\ 0 \\ 1 \end{bmatrix} u$$

and

$$y = [b_n - a_n b_0 \quad b_{n-1} - a_{n-1} b_0 \quad \cdot \quad \cdot \quad b_2 - a_2 b_0 \quad b_1 - a_1 b_0] \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_{n-1} \\ x_n \end{bmatrix} + [b_0] u$$

Jordan Canonical Form

Consider Laplace domain transfer function

$$\begin{aligned}g(s) &= \frac{y(s)}{u(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} \\&= b_0 + \frac{r_1}{(s - \lambda_1)} + \frac{r_2}{(s - \lambda_2)} + \dots + \frac{r_n}{(s - \lambda_n)}\end{aligned}$$

Therefore,

$$\begin{aligned}y(s) &= b_0 u(s) + \frac{r_1 u(s)}{(s - \lambda_1)} + \frac{r_2 u(s)}{(s - \lambda_2)} + \dots + \frac{r_n u(s)}{(s - \lambda_n)} \\&= b_0 u(s) + r_1 x_1(s) + r_2 x_2(s) + \dots + r_n x_n(s)\end{aligned}$$

Where, $x_1, x_2, \dots, x_n$ are considered as the states of the system

Jordan Canonical Form

Therefore, the state equations are

$$\begin{aligned}
 x_1(s) &= \frac{u(s)}{(s - \lambda_1)} & \dot{x}_1 &= \lambda_1 x_1 + u \\
 x_2(s) &= \frac{u(s)}{(s - \lambda_2)} & \dot{x}_2 &= \lambda_2 x_2 + u \\
 &\dots\dots\dots \\
 x_n(s) &= \frac{u(s)}{(s - \lambda_n)} & \dot{x}_n &= \lambda_n x_n + u
 \end{aligned}
 \quad
 \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \vdots \\ \dot{x}_n \end{bmatrix}
 =
 \begin{bmatrix} \lambda_1 & 0 & 0 & \cdot & \cdot & 0 \\ 0 & \lambda_2 & 0 & & & \\ 0 & 0 & \lambda_3 & 0 & \cdot & 0 \\ 0 & 0 & 0 & \lambda_4 & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & \cdot & \cdot & \cdot & \lambda_n \end{bmatrix}
 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \cdot \\ \cdot \\ x_n \end{bmatrix}
 +
 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ \cdot \\ \cdot \\ 1 \end{bmatrix} u$$

and output map

$$y = r_1 x_1 + r_2 x_2 + \dots\dots\dots + r_n x_n + b_0 u$$

What will happen in case of repeated roots?

Jordan Canonical Form (repeated roots)

For repeated roots, the partial fraction expression may be written as

$$\begin{aligned}g(s) &= \frac{y(s)}{u(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} \\&= b_0 + \frac{r_{11}}{(s - \lambda_1)^3} + \frac{r_{12}}{(s - \lambda_1)^2} + \frac{r_{13}}{(s - \lambda_1)} + \frac{r_4}{(s - \lambda_2)} + \dots + \frac{r_n}{(s - \lambda_n)}\end{aligned}$$

Therefore,

$$\begin{aligned}y(s) &= b_0 u(s) + \frac{r_{11} u(s)}{(s - \lambda_1)^3} + \frac{r_{12} u(s)}{(s - \lambda_1)^2} + \frac{r_{13} u(s)}{(s - \lambda_1)} + \frac{r_4 u(s)}{(s - \lambda_2)} + \dots + \frac{r_n u(s)}{(s - \lambda_n)} \\&= b_0 u(s) + r_{11} x_1(s) + r_{12} x_2(s) + r_{13} x_3(s) + r_4 x_4(s) + \dots + r_n x_n(s)\end{aligned}$$

Jordan Canonical Form (repeated roots)

Now the state equations are

$$\begin{aligned}
 x_1(s) &= \frac{x_2(s)}{(s - \lambda_1)} & \dot{x}_1 &= \lambda_1 x_1 + x_2 \\
 x_2(s) &= \frac{x_3(s)}{(s - \lambda_1)} & \dot{x}_2 &= \lambda_1 x_2 + x_3 \\
 x_3(s) &= \frac{u(s)}{(s - \lambda_1)} & \dot{x}_3 &= \lambda_1 x_3 + u \\
 x_4(s) &= \frac{u(s)}{(s - \lambda_2)} & \dot{x}_4 &= \lambda_2 x_4 + u
 \end{aligned}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} \lambda_1 & 1 & 0 & \cdot & \cdot & 0 \\ 0 & \lambda_1 & 1 & & & \\ 0 & 0 & \lambda_1 & 0 & \cdot & 0 \\ 0 & 0 & 0 & \lambda_2 & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & \cdot & \cdot & \cdot & \lambda_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ \cdot \\ \cdot \\ 1 \end{bmatrix} u$$

.....

$$x_n(s) = \frac{u(s)}{(s - \lambda_n)} \quad \dot{x}_n = \lambda_n x_n + u$$

Output map

$$y = r_{11}x_1 + r_{12}x_2 + r_{13}x_3 + r_{14}x_4 + \dots + r_{1n}x_n + b_0u$$

Alternate Canonical form (Toeplitz form)

Consider the transfer function as

$$g(s) = \frac{y(s)}{u(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

$$\text{i.e., } \frac{d^n y}{dt^n} + a_1 \frac{d^{n-1} y}{dt^{n-1}} + \dots + a_{n-1} \frac{dy}{dt} + a_n y = b_0 \frac{d^n u}{dt^n} + b_1 \frac{d^{n-1} u}{dt^{n-1}} + \dots + b_{n-1} \frac{du}{dt} + b_n u$$

Define State equations and output map as the following:

$$y = x_1 + p_0 u$$

$$\dot{x}_1 = x_2 + p_1 u$$

$$\dot{x}_2 = x_3 + p_2 u$$

.....

$$\dot{x}_{n-1} = x_n + p_{n-1} u$$

$$\dot{x}_n = -a_1 x_n - a_2 x_{n-1} - \dots - a_n x_1 + p_n u$$

Alternate Canonical form (Toeplitz form)

From the above definition, we can write

$$y = x_1 + p_0 u$$

$$\dot{y} = \dot{x}_1 + p_0 \dot{u} = x_2 + p_1 u + p_0 \dot{u}$$

$$\ddot{y} = \dot{x}_2 + p_1 \dot{u} + p_0 \ddot{u} = x_3 + p_2 u + p_1 \dot{u} + p_0 \ddot{u}$$

....

$$\frac{d^{n-1}y}{dt^{n-1}} = x_n + p_{n-1}u + p_{n-2}\dot{u} + \dots + p_1 \frac{d^{n-2}u}{dt^{n-2}} + p_0 \frac{d^{n-1}u}{dt^{n-1}}$$

$$\frac{d^n y}{dt^n} = \dot{x}_n + p_{n-1}\dot{u} + p_{n-2}\ddot{u} + \dots + p_1 \frac{d^{n-1}u}{dt^{n-1}} + p_0 \frac{d^n u}{dt^n}$$

$$= -a_1 x_n - a_2 x_{n-1} - \dots - a_n x_1 + p_n u + p_{n-1} \dot{u} + p_{n-2} \ddot{u} + \dots + p_1 \frac{d^{n-1}u}{dt^{n-1}} + p_0 \frac{d^n u}{dt^n}$$

Alternate Canonical form (Toeplitz form)

Therefore,

$$\begin{aligned}
 & \frac{d^n y}{dt^n} + a_1 \frac{d^{n-1} y}{dt^{n-1}} + \dots + a_{n-1} \frac{dy}{dt} + a_n y \\
 &= \left(-a_1 x_n - a_2 x_{n-1} - \dots - a_n x_1 + p_n u + p_{n-1} \dot{u} + p_{n-2} \ddot{u} + \dots + p_1 \frac{d^{n-1} u}{dt^{n-1}} + p_0 \frac{d^n u}{dt^n} \right) \\
 &+ a_1 \left(x_n + p_{n-1} u + p_{n-2} \dot{u} + \dots + p_1 \frac{d^{n-2} u}{dt^{n-2}} + p_0 \frac{d^{n-1} u}{dt^{n-1}} \right) + \dots \\
 &+ a_{n-1} (x_2 + p_1 u + p_0 \dot{u}) + a_n (x_1 + p_0 u) \\
 &= (p_n + a_1 p_{n-1} + \dots + a_{n-1} p_1 + a_n p_0) u + (p_{n-1} + a_1 p_{n-2} + \dots + a_{n-2} p_1 + a_{n-1} p_0) \dot{u} + \dots \\
 &+ (p_1 + a_1 p_0) \frac{d^{n-1} u}{dt^{n-1}} + p_0 \frac{d^n u}{dt^n} \\
 &= b_0 \frac{d^n u}{dt^n} + b_1 \frac{d^{n-1} u}{dt^{n-1}} + \dots + b_{n-1} \frac{du}{dt} + b_n u
 \end{aligned}$$

Alternate Canonical form (Toeplitz form)

Equating the coefficients of $u, \dot{u}, \dots, \frac{d^{n-1}u}{dt^{n-1}}, \frac{d^n u}{dt^n}$

$$b_0 = p_0;$$

$$b_1 = p_1 + a_1 p_0;$$

.....

$$b_{n-1} = p_{n-1} + a_1 p_{n-2} + \dots + a_{n-2} p_1 + a_{n-1} p_0$$

$$b_n = p_n + a_1 p_{n-1} + \dots + a_{n-1} p_1 + a_n p_0$$

In vector-Matrix form,

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 & \cdot & \cdot & 0 & 0 \\ a_1 & 1 & 0 & \cdot & \cdot & 0 & 0 \\ a_2 & a_1 & 1 & \cdot & \cdot & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n-1} & a_{n-2} & \cdot & \cdot & \cdot & 1 & 0 \\ a_n & a_{n-1} & a_{n-2} & \cdot & \cdot & a_1 & 1 \end{bmatrix}}_{\text{Toeplitz Matrix}} \begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ \cdot \\ \cdot \\ p_{n-1} \\ p_n \end{bmatrix} = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ \cdot \\ \cdot \\ b_{n-1} \\ b_n \end{bmatrix}$$

2nd Companion form (Observer Canonical form)

Consider the transfer function

$$g(s) = \frac{y(s)}{u(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{(s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n)}$$

$$\text{i.e., } (s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n) y(s) = (b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n) u(s)$$

Rearranging the terms,

$$s^n [y(s) - b_0 u(s)] + s^{n-1} [a_1 y(s) - b_1 u(s)] + \dots + [a_n y(s) - b_n u(s)] = 0$$

Simplify :

$$y(s) - b_0 u(s) = \frac{1}{s} [b_1 u(s) - a_1 y(s)] + \frac{1}{s^2} [b_2 u(s) - a_2 y(s)] + \dots + \frac{1}{s^n} [b_n u(s) - a_n y(s)]$$

$$y(s) = b_0 u(s) + \frac{1}{s} [b_1 u(s) - a_1 y(s)] + \frac{1}{s^2} [b_2 u(s) - a_2 y(s)] + \dots + \frac{1}{s^n} [b_n u(s) - a_n y(s)]$$

2nd Companion form (Observer Canonical form)

Rearranging the terms,

$$y(s) = b_0 u(s) + \underbrace{\frac{1}{s} \left[\underbrace{[b_1 u(s) - a_1 y(s)] + \frac{1}{s} [b_2 u(s) - a_2 y(s)] + \dots + \frac{1}{s} [b_n u(s) - a_n y(s)]}_{x_2(s)} \right]}_{x_1(s)}$$

The equations now can be written as

$$y = x_1 + b_0 u$$

$$\dot{x}_1 = x_2 - a_1 y + b_1 u = -a_1 x_1 + x_2 + (b_1 - a_1 b_0) u$$

$$\dot{x}_2 = x_3 - a_2 y + b_2 u = -a_2 x_1 + x_3 + (b_2 - a_2 b_0) u$$

.....

.....

$$\dot{x}_{n-1} = x_n - a_{n-1} y + b_{n-1} u = -a_{n-1} x_1 + x_n + (b_{n-1} - a_{n-1} b_0) u$$

$$\dot{x}_n = -a_n y + b_n u = -a_n x_1 + (b_n - a_n b_0) u$$

2nd Companion form (Observer Canonical form)

In vector Matrix form,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -a_1 & 1 & 0 & \cdot & \cdot & 0 \\ -a_2 & 0 & 1 & & & \\ -a_3 & 0 & 0 & 1 & \cdot & 0 \\ \cdot & 0 & 0 & 0 & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ -a_{n-1} & \cdot & \cdot & \cdot & \cdot & 1 \\ -a_n & 0 & 0 & \cdot & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} + \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ b_3 - a_3 b_0 \\ \cdot \\ \cdot \\ b_{n-1} - a_{n-1} b_0 \\ b_n - a_n b_0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & \cdot & \cdot & \cdot & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_{n-1} \\ x_n \end{bmatrix} + b_0 u$$

2nd Companion form (Observer Canonical form)

On the other hand, if we formulate

$$y(s) = b_0 u(s) + \underbrace{\frac{1}{s} \left[\underbrace{[b_1 u(s) - a_1 y(s)] + \frac{1}{s} [b_2 u(s) - a_2 y(s)] + \dots + \frac{1}{s} [b_n u(s) - a_n y(s)]}_{x_1(s)} \right]}_{x_n(s)}$$

Then,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \vdots \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \cdot & \cdot & -a_n \\ 1 & 0 & 0 & 0 & 0 & -a_{n-1} \\ 0 & 1 & 0 & 0 & \cdot & -a_{n-2} \\ 0 & 0 & 1 & 0 & \cdot & -a_{n-3} \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & \cdot & \cdot & \cdot & \cdot & -a_2 \\ 0 & 0 & 0 & \cdot & 1 & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} + \begin{bmatrix} b_n - a_n b_0 \\ b_{n-1} - a_{n-1} b_0 \\ b_{n-2} - a_{n-2} b_0 \\ \cdot \\ \cdot \\ b_2 - a_2 b_0 \\ b_1 - a_1 b_0 \end{bmatrix} u$$